A multicentric phase I/IIA dual-targeted CD19/22 Chimeric Antigen Receptor (CAR)T-cell therapy clinical trial in adult patients with relapsed/refractory CD19-positive B-cell Acute Lymphoblastic Leukemia.
DualCALM-01

CLINICAL TRIAL ON MEDICINAL PRODUCT FOR HUMAN USE

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The study will be carried out in accordance with the protocol, with current good practices and with statutory and regulatory requirements.

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List of abbreviations

AE adverse event
AESI adverse event of Special interest
ALL acute lymphoblastic leukemia
ALT alanine aminotransferase
ALP alkaline phosphatase
aPTT activated partial Thromboplastin time
AST Aspartate Aminotransferase/Serum Glutamic Oxaloacetic Transaminase/SGOT
ANC absolute neutrophil count
AST aspartate aminotransferase
4-1BB Type 2 Transmembrane Glycoprotein (TNF Superfamily, costimulatory domain)
BCR-ABL Philadelphia chromosome
CAR chimeric antigen receptor
CAR+ chimeric antigen receptor positive
CD cluster of differentiation
CI confidence interval
CBC complete blood count
CNS central nervous system
CPF cell processing facility
CR complete remission
CRi complete remission response with incomplete hematologic recovery
CRF case report form
CRS cytokine release syndrome
CSF cerebrospinal fluid
CT computed Tomography
CTCAE common terminology criteria for adverse events
DIC disseminated intravascular coagulation
DLT dose-limiting toxicity
DoR duration of remission
DSMB data safety monitoring board
DVT deep vein thrombosis
eACTTM engineered autologous cell therapy
ECG echocardiogram
ECOG Eastern Cooperative Oncology Group
EEG electroencephalogram
EMA European Medicines Agency
FDA Food and drug Administration
GCP Good clinical practice
G-CSF Granulocyte colony stimulating Factor
GGT Gamma-Glutamyltransferase
GVHD Graft versus Host disease
HBV Hepatitis B Virus
HHV6 Human Herpes virus 6
HIV Human immunodeficiency virus
HLH Hemophagocytic lymphohistiocytosis
HSA human serum albumin
HSV Herpes Simplex Virus
ICANS immune effector cell-associated neurotoxicity syndrome

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ICF informed consent form
ICU intensive care Unit
IHC immunohistochemistry
Ig Immunoglobulin
IL interleukin
IMP investigational medicinal product
IPM investigational product manual
IRB/IEC institutional review board / independant ethics committee
IV Intravenous
LAG-3 lymphocyte activation gene 3
LTFU long-term follow up
LVEF left ventricular ejection fraction
MAS Macrophage activation syndrome
MHC Major Histocompatibility Complex
mITT modified intent to treat
MRD minimal residual disease
MRI Magnetic Resonance Imaging
MTD maximum tolerated dose
NCCN National comprehensive Cancer Network
OS overall survival
PBMC peripheral blood mononuclear cells
PET positron emission tomography
PD progressive disease
Ph+ Philadelphia Chromosome Positive
qPCR quantitative polymerase chain reaction
RBC Red blood cell
RFS relapse free survival
r/r relapsed / refractory
SAE severe adverse event
scFv single chain variable fragment
SMZL splenic marginal zone lymphoma
SOA schedule of assessments
SOC standard of care
SCT stem cell transplant
TIM-3 T-cell Immunoglobulin and mucin-domain containing protein 3
TKI tyrosine kinase inhibitor
TLS tumor lysis syndrome
TCR T cell receptor
TNF tumor necrosis factor
ULN upper limit normal
WBC white blood cell

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1 SUMMARY

Full title	**A multicentric phase I/IIA MB-CART2219.1 Chimeric Antigen Receptor (CAR)T-cell therapy clinical trial in adult patients with relapsed/refractory CD19-positive B-cell Acute Lymphoblastic Leukemia.**
Acronym/reference	**DualCALM-01**
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Sponsor	Assistance Publique – Hôpitaux de Paris Délégation à la Recherche Clinique et à l'Innovation DRCI (Clinical Research and Innovation Department) Hôpital Saint-Louis 1, avenue Claude Vellefaux 75010 PARIS, France
Scientific justification	Relapsed/refractory ALL represents an area of greatest unmet medical need. Second generation autologous anti CD19 CAR-T cells have shown promising results in terms of complete remission rates and relapse-free survival (RFS). Several mechanisms have been shown to lead to the loss of CD19 antigen expression and escape of anti CD19 CAR-T cells therapy. CD22 is observed in more than 90% of BCP-ALL and is already known as an interesting target. Targeting both CD19 and CD22 with our dual CAR-T could enhance CAR-T cells activity and limit tumour escape.
Main objective and primary endpoint	**Main objective** The primary objective of this clinical trial is to answer one key question in each phase: **Phase I:** To determine the **DLT** of MB-CART2219.1, based on safety evaluations conducted within 28 days following infusion. **Phase II:** To evaluate the **overall response rate (ORR**) as measured at day 28 by local assessment.

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	Primary endpoint • Phase I: Dose-Limiting Toxicity (DLT): The primary endpoint for Phase 1 is the evaluation of DLT of MB-CART2219.1 within 28 days after infusion. • Phase II: ORR, defined as the sum of complete remission (CR) and complete remission with incomplete hematologic recovery (CRi) at Day 28, as assessed locally.
Secondary objectives and endpoints	**Secondary objectives:** To evaluate - The safety of MB-CART2219.1; - To evaluate treatment response, duration of response and relapse after treatment with MB-CART2219.1; - To assess phenotype, expansion and persistence of MB-CART2219.1; - To correlate treatment response to CD19 and CD22 expression on leukemic blast cells at baseline; - To evaluate the resources consumed for this strategy during the whole conduction of the study (2 years). **Secondary endpoint:** *Safety Profile*: • Number and detailed description of adverse events (AEs) reported during the 2 years following infusion, graded according to CTCAE v5.0. *Response to CAR-T cells*: • Overall Response Rate (ORR) at Month 3 post-infusion. • Minimal Residual Disease (MRD) Response at Day 28 and Month 3 post-infusion, using flow cytometry or qPCR. *Survival Outcomes*: • DFS: Time from the first complete remission to disease recurrence or progression. • OS: Time from infusion to death from any cause. • DoR: Time from the initial response to disease recurrence or progression, measured up to 2 years post-infusion. *Biological Outcomes:* • Peripheral Blood Monitoring: Quantification of MB-CART2219.1 levels and normal B-cell counts throughout the 2-year follow-up period. • CD19 and CD22 Expression: Measurement of CD19 and CD22 expression levels at baseline and at the time of relapse, if applicable.
Design of the study	This phase I/IIA is a single arm, prospective, open-label study that combines (i) the continual reassessment method (CRM) dose-finding phase I design for evaluation of safety, and (ii) a single-stage phase II design for efficacy evaluation. **Phase 1** dose evaluation/escalation part to define the recommended phase II dose of MB-CART2219.1 in patients aged ≥ 18 years. Dose escalation will be based on cohorts of 3 patients treated at the same dose level. The first cohort will be administered at Dose Level 1. Then, based on Bayesian actualization of DLT probabilities, each subsequent cohort will be treated at the estimated MTD over 4 dose levels, unless dose

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	skipping. In case of DLT at Dose Level 1, the fallback dose of Dose Level 0 will be tested. **Phase 2** to assess the objective clinical response with MB-CART2219.1 at day 28. After the MTD has been defined (phase I), the patients treated at this dose level will be assessed for efficacy
Category	Cat.1 and cat. 2
Population of study participants	Adults with R/R B-cell acute lymphoblastic leukaemia (BCP-ALL)
Inclusion criteria	1. Age ≥ 18 years and < 70 years at inclusion 2. Relapsed or refractory BCP-cell ALL with ≥ 5 % bone marrow blasts meeting either of the following criteria: **a. Second or greater relapse** b. **Any relapse after allogeneic SCT** must be ≥ 3 months from SCT to the time of apheresis c. **Refractory at relapse** (not achieving a CR after one cycle of chemotherapy for relapsed leukemia) Patients must have exhausted relevant authorised, on-label and accessible treatment options 3. Ph+ ALL must be intolerant or failed two lines of TKI therapy. 4. CD19 or CD22 expression demonstrated in BM or peripheral blood by flow cytometry within 3 weeks prior study inclusion. 5. ECOG performance score of 0-2 6. Absolute CD3+ T cell count ≥100/µl 7. No childbearing potential or negative pregnancy test at screening and before chemotherapy in women with childbearing potential 8. Signed and dated informed consent/assent by patients before to start any trial-specific procedure
Exclusion criteria	1. CNS Involvement: Disease involvement of the central nervous system at the time of MB-CART2219.1 lymphodepletion. 2. Recent Medication: One of the following previous medications • Systemic chemotherapies, corticosteroids (with the exception of physiologic replacement dosing), tyrosine kinase inhibitors (TKI), intrathecal therapy within 7 days prior to leukapheresis • Fludarabine/clofarabine or immunosuppressive drugs (e.g., calcineurin inhibitors) and antibodies or antibody-targeted agents (e.g., rituximab, daratumomab, isatuximab, blinatumomab, inotuzumab-ozogamicin) or donor lymphocyte transfusions or radiation therapy within 30 days prior to leukapheresis • Live vaccines within 30 days prior to leukapheresis • Alemtuzumab within 3 months prior to leukapheresis, 3. Administration of other investigational product between apheresis and MB-CART2219.1 infusion, 4. Active clinically significant CNS disease (including but not limited to uncontrolled seizure disorders, cerebrovascular ischemia or haemorrhage, dementia, paralysis),

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	5. Active GVHD or immunosuppression requirement (hydrocortisone > 12 mg/sqm/day or equivalent), 6. Organ failure, defined as a cardiac ejection fraction < 45%; calculated glomerular filtrate rate < 30 mL / min; or bilirubin> 3 times the upper limit of normal, 7. History of uncontrolled cancer for the last 2 years, with the exception of basal cell carcinoma or carcinoma in situ of the cervix, 8. **Viral infection:** Known positive serology for HIV-1 and -2, HTLV-1 and -2, or active hepatitis B or C infection. 9. Positive SARS-Cov2 by PCR on nasal swab, 10. Personal history of autoimmune disease, 11. Contraindication of investigational medicinal product, of auxiliary medicinal products and of trial related procedures 12. Female patients of child-bearing potential not willing to practice a highly effective form of birth control from leukapheresis up to 12 months after MB-CART2219.1 infusion, 13. Male patients of fathering potential not willing to practice a highly effective form of birth control from leukapheresis and up to 12 months after MB-CART2219.1 infusion, 14. Concurrent participation in another interventional trial that could interact with this trial, e.g., CAR T trials.
Investigational medicinal product	Autologous T cells *ex vivo* lentivirally transduced with the tandem (anti-) CD19CD22 chimeric antigen receptor (CAR) construct and expanded (MB-CART2219.1) in a 7 days manufacturing process. In cases where the lymphodepletion regimen cannot be initiated as scheduled, the product may be cryopreserved for later use. Dose escalation: • Dose Level 0: 0.25 x 10^6 CAR-T cells/kg = fallback dose • Dose Level 1: 0.5 x 10^6 CAR-T cells/kg • Dose Level 2: 1.0 x 10^6 CAR-T cells/kg • Dose Level 3: 2.0 x 10^6 CAR-T cells/kg The first cohort will be administered at Dose Level 1. Then, based on Bayesian actualization of DLT probabilities, each subsequent cohort will be treated at the estimated MTD over 4 dose levels, unless dose skipping. In case of DLT at Dose Level 1, the fallback dose of Dose Level 0 will be tested. Marketing authorization: no
Comparator treatment	Not applicable
Interventions added for the study	• Leukapheresis to collect T-cells • Before CAR19/22-T infusion, lymphodepletion course composed of cyclophosphamide (500 mg/sqm/d, D-3 to D-2) and fludarabine (40 mg/sqm/d, D-4 to D-2).
Expected benefits for the participants and for society	Academic production of a new CAR-T cells responding at an actual unmet medical need.

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	While the CAR19/22-T targets two different antigens, it is expected to decrease the risk of relapse that would occur through down regulation of both CD19 and CD22 antigens.
Risks and burdens added by the study	Evaluation of the safety of MB-CART2219.1 a new autologous CAR-T cells. Due to the very advance stages of the diseases considered for inclusion in the trial, the potential benefit of MB-CART2219.1 therapy outweighs the potential risks of the therapy in our study population. Other risks are related to auxiliary medicinal products and to interventions added by the study
Number of participants included	20 patients, 18 patients evaluable
Number of centres	- 5 clinical sites (Paris Saint-Louis, Lyon, Bordeaux, Marseille, Nantes) - 1 ATMP platform (MEARY, Saint-Louis, APHP, Paris)
Duration of the study	Inclusion period: 18 months Participation period (treatment + follow-up): 2 years Total duration: 3.5 years
Number of enrolments expected per site and per month	It is expected that 4 patients will be included by site (0.22 patient/site/month)
Statistical analysis	**Safety/Efficacy analyses** Analysis will be based on the main analysis set, i.e., all patients who are treated with MB-CART2219.1. **Phase 1 data** will be recorded sequentially according to the CRM A working parametric model is first assumed, based on the expected DLT rate at each dose level, assuming a monotonic relationship, with parameter estimation based on recorded data with a noninformative prior. It allows borrowing of information across dose levels along the trial. Before inclusion, model characteristics will be evaluated using a pre-trial simulation study. Analysis will use the dfcrm R package (Cheung YK. Dose finding by the continual reassessment method. Chapman& Hall/CRC Biostatistics Series: Taylor and Francis; 2011). **Phase 2 data** will be analysed in intent-to-treat, based 95% confidence interval of the all the endpoints. **Calculation hypotheses for the number of participants required and the result.** A one-stage Bayesian continual reassessment method (CRM) will be used with cohort size of 3. A maximum sample size of 18 ensures a probability of correct select of 0.5 for the CRM, assuming a 1.8 odds ratio for the true logistic dose-toxicity curve and using a basic arbitrary skeleton with DL2 as the anticipated MTD1. **Method for taking into account missing** Missing data on outcome will not be imputed except for primary outcome of phase 2 data considered as failure if missing.

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Funding sources	INCA (PHRC-K18-148) ; Ligue contre le cancer (AAPTCI2025.LCC)
Study will have a Data Safety Monitoring Board	An independent 3-member data safety monitoring board (DSMB) will be designed and operate according to a separate charter. It will review the trial and decide whether to stop the protocol according to safety events.

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2 SCIENTIFIC JUSTIFICATION FOR THE CLINICAL TRIAL

2.1 Acute lymphoblastic leukemia - Overview

2.1.1.1 Epidemiology

Acute lymphoblastic leukaemia (ALL) in adults is a rare but aggressive malignancy characterized by the uncontrolled proliferation of immature lymphoid cells, typically affecting the **bone marrow and peripheral blood**. Although ALL is more commonly associated with children, where the prognosis is generally favourable, the outlook for adults is significantly worse, especially in the relapsed/refractory (R/R) setting.[1]

Each year, approximately **400 new cases** of ALL are diagnosed in France among adults,[2] with similar incidence rates across Europe. Adult ALL typically presents in patients between 18 and 60 years of age, with a slight male predominance.

B-cell precursor ALL (BCP-ALL) accounts for approximately **75-80% of adult ALL cases**, while the remaining cases involve T-cell lineage. ALL is characterized by the overproduction of immature lymphoid cells, which fail to differentiate and infiltrate the bone marrow, blood, and other extramedullary sites like the **central nervous system (CNS)**, liver, and spleen.[3]

The most common subtype in adults is **Philadelphia chromosome-positive (Ph+) ALL**, a condition caused by the ***BCR::ABL1*** fusion protein, resulting from the **t(9;22)** translocation. Ph+ ALL is historically associated with poor outcomes. The introduction of **tyrosine kinase inhibitors (TKIs)** has dramatically improved remission rates and long-term outcomes.[4,5] The prognosis for Ph-negative ALL in adults has dramatically improved thanks to the use of more intensive paediatric-inspired regimen.[6,7] Increasing age remains a strong limitation of these intensive strategies due to increased toxicities and treatment-related mortality.

2.1.1.2 Prognostic Factors and Survival Rates

The **prognosis** for adults with ALL, especially those with relapsed/refractory disease, is significantly worse than in paediatric populations. While children with ALL can achieve **5-year overall survival rates of up to 90%**, adult patients face a **5-year survival rate of only 40-70%**.[8] Age is one of the most significant prognostic factors; older adults (above 40-50 years) tend to have poorer outcomes due to more aggressive disease biology, comorbidities, and decreased tolerance to intensive chemotherapy.

Cytogenetic abnormalities also play a major role in prognosis. Besides **Ph+ ALL**, other high-risk cytogenetic abnormalities include hypodiploidy**,** KMT2A rearrangements, or Ph-like profile.[1,3] Patients with these high-risk genetic features tend to relapse more frequently and are less responsive to standard therapies.

2.1.1.3 Relapsed/Refractory (R/R) B-ALL in Adults

For adults with **R/R B-ALL**, the prognosis is extremely poor, with a median survival of **less than 6 months**. These patients typically do not respond well to conventional chemotherapy, and their outcomes have not significantly improved over the past several decades. Standard chemotherapy regimens result in **complete remission (CR)** rates of less than **20-30%** for patients in second or third relapse, and the duration of remission is often short.[8]

Patients with **primary refractory disease**—those who fail to achieve remission after two cycles of induction chemotherapy—have especially poor outcomes. For these patients, **allogeneic stem cell transplantation (allo-SCT)** is often the only potential curative option, but many are not eligible due to advanced disease, comorbidities, or lack of suitable donors.

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2.1.1.4 Advances in Immunotherapy for R/R B-ALL

Recent advances in **immunotherapy** and **targeted therapies** have opened new avenues for treating R/R B-ALL in adults, offering hope where traditional chemotherapy has failed. Immunotherapies aim to harness or enhance the body's immune response to eliminate cancer cells.[9]

Monoclonal Antibodies:

- **Blinatumomab**, a bispecific T-cell engager (BiTE) that targets **CD19** on B-cells and **CD3** on T-cells, has significantly improved outcomes for patients with R/R B-ALL. Blinatumomab has shown promising results, achieving remission in patients who have relapsed or are refractory to conventional therapies.[10]
- **Inotuzumab ozogamicin**, a CD22-targeted antibody-drug conjugate, has also demonstrated efficacy in inducing remission in adult patients with R/R B-ALL.[11]

CAR-T Cell Therapy:

- The most groundbreaking development in the treatment of R/R B-ALL is the use of **chimeric antigen receptor T-cell (CAR-T) therapy**. This treatment involves genetically modifying a patient's T-cells to express a CAR that specifically targets **CD19** on B-cells, including leukemic blasts.
- CAR-T cell therapy has shown **high complete remission rates (50-90%)** in heavily pre-treated adults with R/R B-ALL, including those who have relapsed after allo-SCT or are ineligible for transplant.
- Two CAR T-cell are currently approved for the treatment of R/R B-ALL in children and young adults (tisagenlecleucel) and adults (brexucabtagene autoleucel).[9,12,13]
- However, CAR-T therapy is associated with significant toxicity, particularly **cytokine release syndrome (CRS)** and **immune effector cell-associated neurotoxicity syndrome (ICANS)**, which require careful monitoring and management.

One major challenge after the use of blinatumomab or CD19 CAR T-cells is the emergence of **CD19-negative relapse** due to the loss or downregulation of CD19 on leukemic cells. This **antigen escape** mechanism allows the disease to evade immunotherapy, especially when these therapies are used sequentially.[14,15]

2.1.1.5 The Need for Novel Treatments

Despite advances in immunotherapy, **relapsed/refractory B-ALL in adults** remains an area of unmet clinical need. The use of CAR-T cells targeting both **CD19** and **CD22** may provide a more durable response and reduce the likelihood of antigen escape, a common mechanism of relapse in single-target therapies.

Given the poor prognosis for adults with R/R B-ALL and the high rates of relapse after allo-SCT and chemotherapy, novel therapies like **dual-target CAR-T cells** represent a critical development in the treatment landscape. These therapies aim to improve survival, achieve minimal residual disease (MRD)-negative remissions, and provide a bridge to transplantation or long-term disease control.

2.2 Summary of relevant pre-clinical experiments and clinical trials

Immunotherapy plays an increasing role in the treatment of haematological malignancies, from the first approval of rituximab about 20 years ago to the recent authorizations delivered to the chimeric antigen receptor T cells in B-cell acute leukaemia (BCP-ALL) and diffuse large B-cell non-Hodgkin lymphoma (DLBCL).[12,16–18]

Interestingly, **the CAR-T cell strategy** has been developed about 25 years ago with the aim to give the capacity to T-cell to recognize tumor associated surface antigen and to bypass the major histocompatibility complex.[19,20] The first-generation CAR was typically composed of an extracellular antibody single-chain fragment (scFv) fused to a transmembrane domain, and the intracellular signalling domain of the T-cell receptor CD3zeta chain. To improve the properties of the CAR-T, second

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generation CAR have included intracellular domain from different costimulatory molecules like CD28, 4-1BB (CD137) or OX40 (CD134).[21]

Autologous CAR-T cells are engineered from patient lymphocytes. In B-cell malignancies including BCP-ALL and DLBCL, second generation CAR-T that recognize CD19 antigen have shown promising results in terms of complete remission rates and relapse/progression-free survival (RFS/PFS). The ELIANA study enrolled 92 children and young adults with relapsed or refractory (R/R) BCP-ALL treated with the anti-CD19/4-1BB/CD3dzeta tisagenlecleucel.[12] Among the 75 evaluable patients, 61 (81%; 95% CI, 71 to 89) went into complete remission with or without complete hematologic recovery. All patients who responded experienced a complete minimal residual disease (MRD) response and the RFS was 59% (95% CI, 41 to 73) at 12 months. Long-term follow-up shows a 36-month OS of 62.8%. In the adult Memorial Sloan Kettering Cancer Center (MSKCC) study, 83 adult patients with R/R BCP-ALL were enrolled to receive an anti-CD19/CD28/CD3dzeta autologous CAR-T.[17] Of the 53 patients who received the CAR-T, 44 (83%; 95% CI, 70 to 92) had a complete remission. A MRD complete remission rate was observed in 32/48 evaluable patients (67%; 95% CI, 52 to 80). The median event-free survival was 6.1 months (95% CI, 5.0 to 11.5) and the median overall survival was 12.9 months (95% CI, 8.7 to 23.4). Data recently updated from ZUMA-3 study show a median OS of 47.0 months after 3 years of follow-up in responders[22].

Furthermore, phase I clinical trials and retrospective studies have reinforced the potential of bispecific CD19/CD22 CAR-T cell therapies in such patient populations. Notably, the clinical study assessing [23], CD19/CD22-41BBz CAR-T cells manufactured with vector provided by Miltenyi Biotec®, has displayed an ORR of approximately 80–85%, including a CR rate of approximately 80% (predominantly MRD-negative), in children and young adults with high-risk R/R B-ALL.

A non-randomized comparative clinical study in R/R B-ALL patients[24], comparing single-target CD19 CAR-T therapy to CD19/CD22 CAR-T therapy showed similarly high initial response rates (ORR ~85–90%, CR ~70–80%); however, dual-target approaches were associated with a reduced incidence of CD19-negative relapse.

More recently, a phase II trial with autologous dual-targeting CD19/CD22 CAR-T cells, the observed CR rate and MRD negativity were 91.7% and 83.3% respectively, in treated adult patients with high-risk Philadelphia chromosome–negative B-ALL, furthermore strengthening the interest for such dual-target therapy in adult B-ALL patients[25].

Despite these impressive results, **many questions remain to be addressed in the field of CAR-T therapy**. The first large phase II studies clearly demonstrated the feasibility of centralized CAR-T manufacturing.[12,16] However the rate of patients that were infused remains low when compared to the patients included, especially in rapid progressive disease like ALL. In the ELIANA and MSKCC studies, 82% and 65% of R/R BCP-ALL patients were infused respectively, with 4-8% of death before infusion and 2-8% manufacturing issues.[12,16] Specific serious adverse events include potentially life-threatening CRS and neurotoxicity. A severe CRS (Grade 3-4) is observed in about 25-50% of patients with BCP-ALL, less frequently in R/R lymphoma patients and seems to be related to many factors, including the CAR-T dose, the tumour burden and the CAR costimulatory domain.[26] Many strategies are being developed to increase the safety of CAR-T, including a greater tumour burden reduction before CAR-T infusion and intra-patient dose escalation. The ability of tumours to escape by the loss of CD19 antigen expression is a real concern, with a specific focus on patients that were pre-exposed to anti-CD19 immunotherapy like blinatumomab. Many mechanisms have been shown to lead to the loss of CD19 antigen expression, including the loss of the entire CD19 encoding gene, mutations leading to alternative splicing and the expression of truncated CD19 protein, or the lineage switch from B-ALL to AML.[27,28] Finally, the cost of these drugs recently approved by the FDA, are expected to be a significant burden on our health care system.

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In this phase I/II clinical trial, we aim to demonstrate our ability to produce and infuse an academic CAR-Ts in a multicentric, safe and efficient way, with regards to resources consumed. In adult patients with B-cell malignancies including relapsed/refractory (R/R) BCP-ALL including those pre-exposed to anti-CD19 bispecific and/or CAR-T therapy, we will propose to enter this phase I/II trial to evaluate the safety and efficacy of dual-targeted CD19/22 CAR-T therapy. The primary objective of this study will be to evaluate the safety of the CAR-T administrated with a risk mitigation strategy based on intra-patient dose-escalation. Finally, we aim to estimate the global cost of this academic approach, associated to the healthcare resources consumed by health system for this complex patient pathway.

2.3 Description of the population to be studied and justification for the choice of participants

The target population for this clinical trial comprises adult patients diagnosed with relapsed or refractory (R/R) B-cell precursor acute lymphoblastic leukaemia (BCP-ALL). This population faces limited treatment options and poor prognosis, particularly following failure of conventional therapies. Clinical data from pivotal trials and real-world studies on CAR-T cell therapies demonstrate promising efficacy in this cohort, making them ideal candidates for this investigational treatment.

Specifically, participants in **this clinical trial will include**:

- Adult patients who are chemo-refractory, meaning they have not achieved or maintained a response following prior systemic therapies.
- Patients who have relapsed after undergoing allogeneic stem cell transplantation (SCT), with a sufficient post-SCT interval to ensure patient safety.
- Patients deemed ineligible for SCT, either due to age, comorbidities, or other contraindications, will also be included, addressing the unmet medical need in these individuals.

Given that **this clinical trial evaluates a dual-target CAR-T therapy directed against CD19 and CD22**, only patients with CD19-positive BCP-ALL will be enrolled. These surface antigens are critical for the therapeutic action of the investigational product, as loss of expression may result in a lack of efficacy. Patients with T-cell lineage ALL will be excluded, as their malignancies do not express CD19 or CD22, and would therefore not respond to this CAR-T therapy.

2.4 Identification and description of the investigational medication (see investigational brochure for more details)

General Identification

MB-CART2219.1 is classified as an advanced therapeutic medicinal product (ATMP) developed by Miltenyi Biomedicine GmbH. It specifically targets CD19- and CD22-expressing cells and is designed to treat hematologic malignancies such as B-cell acute lymphoblastic leukaemia (B-ALL) and diffuse large B-cell lymphoma (DLBCL). The medication comprises a suspension of non-cryopreserved, CD4/CD8-enriched T cells derived from an autologous, non-mobilized leukapheresis.

Pharmaceutical Composition

The MB-CART2219.1 cellular product is composed of autologous T cells that have been genetically engineered via a lentiviral vector to express a third-generation tandem chimeric antigen receptor (CAR) targeting both CD19 and CD22 antigens. The CAR construct incorporates:

- An extracellular antigen-binding domain composed of human CD22-binding single-chain variable fragment (scFv) 16p17, linked via a glycine-serine flexible linker to a human CD19-binding scFv M19217-1.
- A CD28 transmembrane domain linked sequentially to CD28 and CD137 (4-1BB) costimulatory domains.

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- A CD3ζ intracellular signaling domain essential for T-cell activation.

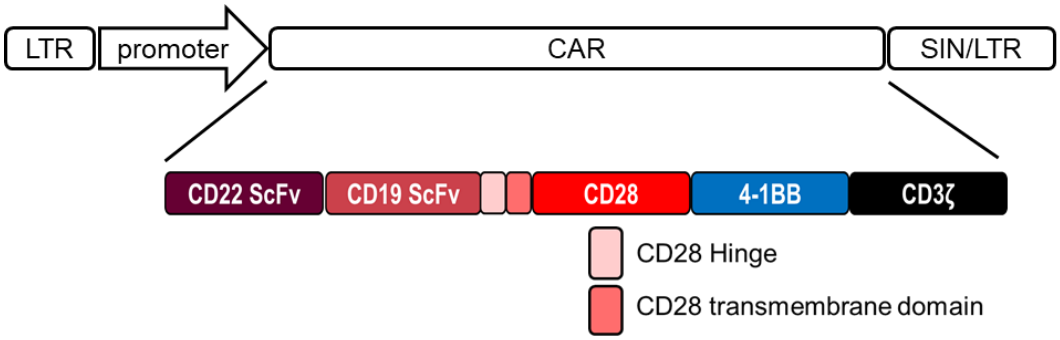


Figure 1: The (anti-) CD2219 CAR sequence (Viral genome on top, CAR structure at the bottom)

The genetically modified cells are suspended in CliniMACS Formulation Solution, which includes human serum albumin (HSA) and various electrolytes such as sodium chloride, sodium gluconate, potassium chloride, and magnesium chloride hexahydrate.

Manufacturing Process

The MB-CART2219.1 product is manufactured using the automated, closed-system CliniMACS Prodigy® device. The process involves:

- Selection and enrichment of autologous CD4/CD8 T cells.
- Genetic modification of T cells using a self-inactivating lentiviral vector carrying the CD2219 CAR construct.
- A 7-day manufacturing process, where transduction efficiency, cell viability, cell count, cellular composition, T-cell differentiation, and exhaustion profiles are closely monitored.

The final product demonstrates high purity and viability, typically composed predominantly of early-differentiated memory T-cells, mainly stem-cell memory-like T cells (TSCM) and central memory T cells (TCM).

Formulation

MB-CART2219.1 is presented as a cellular dispersion for intravenous infusion, in a formulating solution composed of NaCl 0.9% and 0.5% of HAS into in a 600mL administration bag (Macopharma, VSE4002XA).
MB-CART2219.1 is administered as a fresh cell therapy infusion, when possible, to ensure maximal cell viability and potency. In cases where immediate infusion is not feasible, the product may be cryopreserved and infused at a later time.

Summary of Nonclinical Studies

In preclinical studies, MB-CART2219.1 demonstrated substantial efficacy through multiple assays. *In vitro* tests revealed robust and specific cytolytic activity against CD19 and CD22-expressing cell lines, with absolute cytolytic activities exceeding 50% at effector-to-target ratios as low as 2.5:1 (Figure 2). Additionally, CAR T cells efficiently secreted significant amounts of critical cytokines, notably interferon-gamma (IFN-γ), interleukin-2 (IL-2), tumor necrosis factor-alpha (TNF-α), and granulocyte-macrophage colony-stimulating factor (GM-CSF), upon exposure to target antigens (Figure 3).

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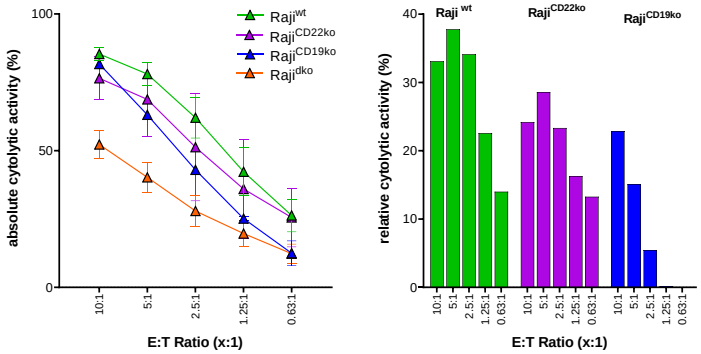


Figure 2. A: Percentages of absolute specific (RajiWT RajiCD22ko RajiCD19ko) and unspecific (Rajidko) cytolytic activity and B: relative cytolytic activity at which unspecific absolute lysis by Rajidko is subtracted from specific lysis detected RajiWT RajiCD22ko

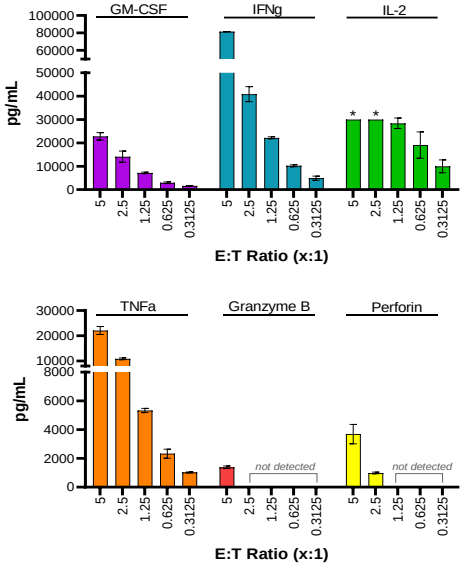


Figure 3. MB-CART2219.1 DP *in vitro* cytokine response

In vivo efficacy studies were conducted using immune-deficient NSG mice (NOD scid gamma (NOD.Cg-Prkdcscid Il2rgtm1Wjl/SzJ)) bearing CD19/CD22-positive tumors (Raji cells, Figure 4). These experiments demonstrated potent, dose-dependent antitumor responses, resulting in tumor regression or complete elimination (Figure 5). Notably, a single administration of MB-CART2219.1 resulted in substantial therapeutic benefits, with higher doses correlating with more rapid and complete tumor control. Persistence of MB-CART2219.1 cells was consistently documented in blood, spleen, and bone marrow at the conclusion of the studies. Importantly, these nonclinical studies reported no significant toxicities, indicating an acceptable safety profile for progression to clinical trials.

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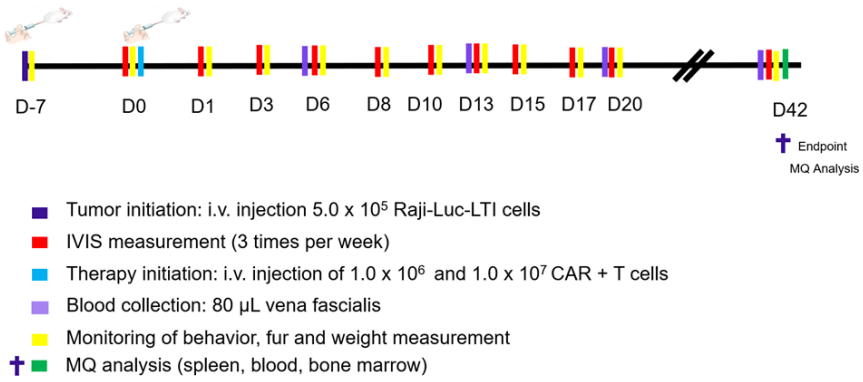


Figure 4.: Overview of the *in vivo* efficacy study workflow

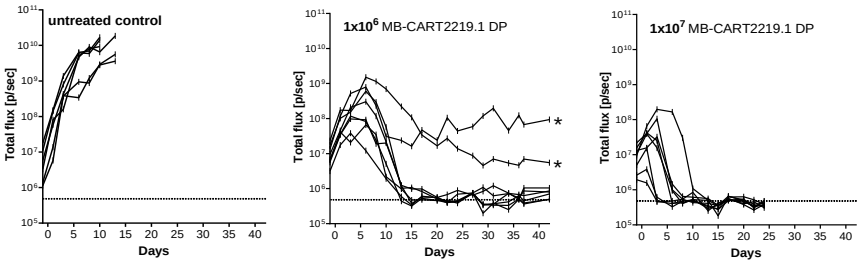


Figure 5. MB-CART2219.1 DP in vivo efficacy (tumor burden, total flux)

2.5 Description and justification of the dosage, route of administration, administration schedule and treatment duration

MB-CART2219.1 will be administered via a single intravenous (IV) infusion following a standard lymphodepleting regimen. The choice of IV administration aligns with established protocols for CAR-T cell therapies.

Justification of Dosage:

The dose escalation strategy for MB-CART2219.1 is based on well-established principles of CAR-T cell therapy to ensure patient safety while identifying the optimal therapeutic dose. The dosing schedule will include the following levels, adjusted according to patient body weight (cells/kg):

- Dose Level 0: 0.25 x 10^6 CAR-T cells/kg
- Dose Level 1: 0.5 x 10^6 CAR-T cells/kg
- Dose Level 2: 1.0 x 10^6 CAR-T cells/kg
- Dose Level 3: 2.0 x 10^6 CAR-T cells/kg

This approach will allow for careful monitoring of DLTs and other adverse events, as well as the determination of the MTD. Each cohort will consist of three patients, with an interval of at least one month between the first and second patients in each dose group. The first cohort will be administered at Dose Level 1. Dose escalation will proceed based on recommendations from the algorithm detailed in section 12.

Route of Administration:

The MB-CART2219.1 product will be administered via a single IV infusion. This route is standard for CAR-T, enabling the engineered T cells to circulate and locate target malignant B-cells expressing CD19 and/or CD22.

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Lymphodepletion Regimen:

Prior to the CAR-T infusion, all patients will undergo a lymphodepleting regimen. This preparative regimen includes:

- Fludarabine: 40 mg/m²/day intravenously on Days -4, -3, and -2.
- Cyclophosphamide: 500 mg/m²/day intravenously on Days -3 and -2.

This lymphodepleting chemotherapy is critical to enhance the persistence and expansion of the infused CAR-T cells by creating a favourable environment in the patient's immune system. Patients with reduced renal function (creatinine clearance between 30 and 70 mL/min/1.73 m²) will receive adjusted fludarabine doses and will be closely monitored for renal function throughout treatment.

Manufacturing and Scheduling:

The MB-CART2219.1 product is manufactured from autologous T cells collected via leukapheresis. The manufacturing process, which spans seven days, includes lentiviral transduction to insert the CD19/CD22 CAR construct into the patient's T cells, followed by cell expansion. The final product is as fresh cells when possible. In cases where the lymphodepletion regimen cannot be initiated as scheduled, the product may be cryopreserved for later use.

Bridging Chemotherapy:

For patients with a high tumour burden, defined by either >25% leukemic blasts in the marrow or ≥10 G/L blasts in peripheral blood, optional bridging chemotherapy may be administered between leukapheresis and lymphodepletion. This is designed to reduce tumour burden and prevent complications such as TLS, CRS, and ICANS. Bridging therapy will be administered at the discretion of the treating clinician, based on the patient's history, clinical condition, and disease burden.

Treatment Duration and Follow-Up:

Patients will receive a single infusion of MB-CART2219.1 on Day 0 following completion of the lymphodepletion regimen. Close clinical and biological monitoring will be performed for a minimum of 28 days post-infusion to assess for acute toxicities, such as CRS and immune effector cell-associated neurotoxicity syndrome (ICANS).

Post-treatment, patients will continue regular follow-up visits at Days 28, and Months 3, 6, 12, 18, and 24 to monitor:

- Bone marrow disease status
- CAR-T cell persistance (MB-CART2219.1 quantification)
- B-cell aplasia
- Response to therapy (e.g., remission status)

The duration of follow-up will extend to two years, with continued monitoring for safety and efficacy. This schedule allows for the comprehensive assessment of both short-term and sustained responses to the therapy, as well as identification of any later-onset adverse events.

According to the applicable regulations, patients should be followed for 15 years after the IMP injection.

2.6 Summary of the known and foreseeable benefits and risks for the research participants

Benefits:

Participation in this clinical trial offers potential individual benefits for patients with R/R BCP-ALL, a population with limited treatment options and poor prognosis. The investigational product, **MB-CART2219.1**, is designed to target both CD19 and CD22 antigens, potentially enhancing the efficacy of CAR-T cell therapy by reducing the risk of relapse due to antigen loss. For patients who have exhausted

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standard therapies, this novel dual-target CAR-T approach could result in prolonged remission, improved survival outcomes, and potentially curative responses.

On a broader scale, this study contributes to advancing knowledge about dual-target CAR-T therapies and their effectiveness in patients who have relapsed after or are ineligible for allogeneic stem cell transplantation (SCT). The data generated from this research may lead to better treatment strategies for similar patient populations in the future.

Foreseeable risks:

The risks associated with participation in this clinical trial include those related to the administration of the IMP, **MB-CART2219.1**, as well as risks from the lymphodepleting regimen, additional procedures including leukapheresis, and long-term monitoring.

1. **Risks from MB-CART2219.1:**
    - **Cytokine Release Syndrome (CRS):** CRS is a common and potentially severe adverse effect of CAR-T cell therapies, characterized by fever, hypotension, hypoxia, and multi-organ dysfunction. Severe cases may require intensive care and immunosuppressive treatment (e.g., tocilizumab or corticosteroids).
    - **Immune Effector Cell-Associated Neurotoxicity Syndrome (ICANS):** Neurotoxicity, including confusion, seizures, or encephalopathy, may occur following CAR-T infusion. This risk will be closely monitored and managed per ASTCT guidelines.
    - **B-cell aplasia and risk of infection:** The targeting of CD19 and CD22 may result in long-term B-cell aplasia, which increases susceptibility to infections. Prophylactic immunoglobulin replacement therapy may be required.
    - **Tumor lysis syndrome (TLS):** Patients with a high disease burden may develop TLS, which can cause severe electrolyte imbalances and renal failure. This risk is managed with appropriate prophylaxis and monitoring.
2. **Risks from the Lymphodepleting regimen:**
    - The lymphodepleting chemotherapy (fludarabine and cyclophosphamide) administered prior to CAR-T cell infusion is associated with transient cytopenias (neutropenia, anemia, thrombocytopenia), infection risk, and organ toxicity (e.g., renal or hepatic impairment).
3. **Procedural and monitoring risks:**
    - **Leukapheresis:** This procedure, used to collect T cells, is generally well-tolerated but carries minor risks such as fatigue, dizziness, or complications related to central venous access.
    - **Frequent blood sampling and imaging:** The follow-up schedule includes regular blood tests and imaging studies to assess disease status, which may impose an additional burden on participants in terms of time and discomfort.
4. **Long-term risks:**
    - Patients will be monitored for potential delayed adverse events, such as the development of secondary malignancies including CAR+ lymphoproliferations, or autoimmune conditions, which are possible but rare long-term risks associated with gene-modified therapies like CAR-T cells.

Risk/Benefit Assessment:

The risk/benefit ratio for this clinical trial is favourable given the patient population targeted. Individuals with relapsed/refractory BCP-ALL face a high likelihood of disease progression and mortality, with limited treatment options available after failure of standard therapies. While the investigational treatment carries known risks, including CRS, neurotoxicity, and B-cell aplasia, these

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risks are justifiable given the potential for achieving remission or extending survival in a population with few alternatives.

A comprehensive safety monitoring plan is in place to mitigate risks, including the use of established protocols for managing CRS and ICANS, regular patient assessments, and the involvement of a Data Safety Monitoring Board (DSMB) to oversee safety data and make adjustments as needed.

All centers participating in the clinical trial are authorized for their CAR-T activities and are certified as institutions conducting research involving human participants (LRIPH).[1,2]

[1] Arrêté du 19 mai 2021 limitant l'utilisation de médicaments de thérapie innovante à base de lymphocytes T génétiquement modifiés dits CAR-T Cells autologues à certains établissements de santé en application des dispositions de l'article L. 1151-1 du code de la santé publique

[2] https://www.legifrance.gouv.fr/codes/article_lc/LEGIARTI000046125746/2022-07-31/

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3 OBJECTIVES

3.1 Primary objective

The primary objective of this clinical trial is to answer one key question in each phase:

Phase I:
To determine the **DLT** of MB-CART2219.1, based on safety evaluations conducted within 28 days following infusion.

Phase II:
To evaluate the **overall response rate (ORR**) as measured at day 28 by local assessment.

3.2 Secondary objectives

The secondary objectives of this clinical trial are to evaluate the following:
- The safety of MB-CART2219.1;
- To evaluate treatment response, duration of response and relapse after treatment with MB-CART2219.1;
- To assess phenotype, expansion and persistence of MB-CART2219.1;
- To correlate treatment response to CD19 and CD22 expression on leukemic blast cells at baseline;
- To evaluate the resources consumed for this strategy during the whole conduction of the study (2 years).

4 STUDY DESIGN

4.1 Study endpoints

4.1.1 Primary endpoint

Phase I:

Dose-Limiting Toxicity (DLT): The primary endpoint for Phase 1 is the evaluation of DLT of MB-CART2219.1 within 28 days after infusion. The DLT is defined as any of the following adverse events (AEs) related to the investigational medicinal product (IMP):

- **Any non-haematological, grade 4 or higher, clinically relevant adverse event (AE), with the following exceptions**:
  - Fever or infection'
  - Laboratory abnormalities judged not to be clinically relevant
  - Events that can be promptly controlled by appropriate medical management
  - TLS grade ≥ 4 resolved within 7 days (see below)
- **Immune effector cell-associated neurotoxicity syndrome (ICANS)**
  - Any grade ≥ 4
  - Grade 3 that fails to resolve to grade ≤ 2 within 72 hours despite appropriate therapy
- **Tumour lysis syndrome (TLS)**
  - Any grade ≥ 4 that fails to resolve within 7 days
- **Cytokine release syndrome (CRS)**
  - Any grade ≥ 4
  - Grade 3 that fails to resolve to grade ≤ 2 within 72 hours despite appropriate therapy

All AEs will be graded according to CTCAE v6.0, with the exception of CRS and ICANS, which will be graded according to the American Society for Transplantation and Cellular Therapy (ASTCT) scale.

Phase II:

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ORR: The primary endpoint for Phase 2 is the ORR, defined as the sum of complete remission (CR) and complete remission with incomplete hematologic recovery (CRi) at Day 28, as assessed locally.

4.1.2 Secondary endpoints

Safety Profile:

- Number and detailed description of adverse events (AEs) reported during the **2 years** following infusion, graded according to **CTCAE v6.0**.

Response to CAR-T cells:

- Overall Response Rate (ORR) at Month 3 post-infusion.
- Minimal Residual Disease (MRD) Response at Day 28 and Month 3 post-infusion, using flow cytometry or qPCR.

Survival Outcomes:

- DFS: Time from the first complete remission to disease recurrence or progression.
- OS: Time from infusion to death from any cause.
- DoR: Time from the initial response to disease recurrence or progression, measured up to 2 years post-infusion.

Biological Outcomes:

- Peripheral Blood Monitoring: Quantification of MB-CART2219.1 levels and normal B-cell counts throughout the 2-year follow-up period.
- CD19 and CD22 Expression: Measurement of CD19 and CD22 expression levels at baseline and at the time of relapse, if applicable

4.2 Description of research methodology

4.2.1 Design of the clinical trial

This Phase I/IIA clinical trial is designed as a single-arm, prospective, open-label clinical trial to evaluate the safety and efficacy of MB-CART2219.1, a dual-targeted CD19/22 chimeric antigen receptor (CAR) T-cell therapy, in adult patients with relapsed/refractory (R/R) CD19-positive B-cell acute lymphoblastic leukaemia (BCP-ALL).

Phase I: Dose-Escalation clinical trial

- **Objective:** To determine the MTD and recommended dose level for Phase II (RDP2) of MB-CART2219.1.
- **Design:** The dose-escalation phase will utilize the continual reassessment method (CRM) to evaluate safety. This phase will include cohorts of 3 each, treated sequentially at escalating dose levels of MB-CART2219.1. The first cohort will be administered at Dose Level 1.
    - **Dose Levels:**
        - **Dose Level 0:** 0.25 x 10^6 CAR-T cells/kg
        - **Dose Level 1:** 0.5 x 10^6 CAR-T cells/kg
        - **Dose Level 2:** 1.0 x 10^6 CAR-T cells/kg
        - **Dose Level 3:** 2.0 x 10^6 CAR-T cells/kg
    - **Safety Monitoring:** An interval of at least one month between the treatment of the first and second patient in each dose cohort is mandatory. Escalation to the next dose level will be decided after at least three patients have been treated at a given dose depending on the algorithm. In case of DLT at Dose Level 1, the fallback dose, Dose Level 0 will be tested.

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Phase II: Efficacy Evaluation

- **Objective:** To assess the clinical efficacy of MB-CART2219.1 at the MTD identified in Phase I.
- **Design:** This phase is a roll-over study including all patients treated at the MTD in phase 1 among the 18 evaluable patients included in phase 1. Overall response rate at day 28 will be described with its 95% confidence interval

4.2.2 Number of participating sites

Recruitment centres

This clinical trial will be conducted across five clinical sites in France, each selected based on their expertise in haematology and their capability to manage and monitor advanced therapy medicinal products (ATMPs), including CAR-T cell therapies. All sites are accredited for CAR T-cell therapy and for first-in-human clinical trials.

N°	Name	Surname	Town	Country	Expected recruitment/month	Total
1	BOISSEL	Nicolas	Paris	France	0.22	4
2	BALSAT	Marie	Lyon	France	0.22	4
3	LEGAY	Thibault	Bordeaux	France	0.22	4
4	HICHERI	Yosr	Marseille	France	0.22	4
5	CHEVALLIER	Patrice	Nantes	France	0.22	4

Centralized Manufacturing Facility

The Paris Saint-Louis Hospital ATMP Platform MEARY will serve as the centralized facility for the manufacturing of the MB-CART2219.1 CAR-T cells. Leukapheresis products from each clinical site will be sent to this platform, where the T cells will be processed, expanded, and prepared for infusion. Depending on the clinical needs, the CAR-T cells will be administered either fresh or cryopreserved for later use.

4.2.3 Identification of participants

Patient identification and recruitment:

- Patients will be identified and recruited through classical referral pathways established within each participating centre and national cooperative groups (e.g., GRAALL, SFCE).
- Each participating site will utilize standardized protocols for patient identification, leukapheresis, and follow-up to ensure consistency and quality across the study.

Patient identification:

- Each participant in this research will be assigned a unique identifier formatted as follows: Site number (3 digits) - Sequential enrolment number for the site (4 digits).
- This identifier will be used throughout the study for all patient records and communications to maintain confidentiality and data integrity.

5 IMPLEMENTATION OF THE STUDY

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5.1 Patient journey and IMP manufacturing

The process begins with the **screening visit**, where the patient's eligibility is confirmed, followed by the **leukapheresis** to collect the patient's T cells for CAR-T manufacturing.
Leukapheresis product is frozen and sent to the manufacturing site (MEARY, Saint-Louis Hospital). A minimum of $50x10^6$/mm3 CD4+CD8+ is needed to start the manufacturing. A second day of leukapheresis can be done if needed.

After leukapheresis, patients may receive **optional bridging therapy** depending on their disease burden. If bridging therapy is administered, the start of the CAR-T manufacturing process is postponed until the patient is ready for lymphodepletion.

To ensure that the patient's treatment journey aligns with the IMP manufacturing, the clinical site and the manufacturing site must coordinate through a **4-step synchronization process** (figure 6):

1. **"Go" for manufacturing:** The clinical site informs the manufacturing site that the patient is ready to begin the lymphodepletion regimen. **A maximum of 14 days may be required between this date and the availability of the IMP (D0). Due to the manufacturing process and operator requirement manufacturing can start either on Monday or on Thursday.**
2. **Theoretical IMP infusion date:** The manufacturing site proposes a theoretical infusion date (**D0**) based on their production schedule, which must be confirmed by the clinical site (clinicians and pharmacists). The manufacturing process begins from the frozen leukapheresis product according to this date.
3. **"Go" for lymphodepletion:** On the first day of the manufacturing process (D-8 before infusion), the manufacturing site confirms the infusion date (D0). This confirmation allows the clinical site to start lymphodepletion on D-4.
4. **"Go" for fresh IMP infusion:** Once the lymphodepletion is completed, the clinical site confirms the infusion date (**D0**). If the patient could not start lymphodepletion, the IMP is frozen for a postponed infusion.

To ensure proper coordination, a **Coordination Form** (addendum X) is submitted by both sites at each step of the process: for the "Go for Manufacturing", the "Theoretical IMP infusion date", the "Go for Lymphodepletion", and the "Go for fresh IMP infusion". The form should be first sent to Meary at caroline.deoliveira@aphp.fr, helene.boucher@aphp.fr, jean-roch.fabreguettes@aphp.fr The form should be shared by the local pharmacists in charge of IMP reception and infusion.

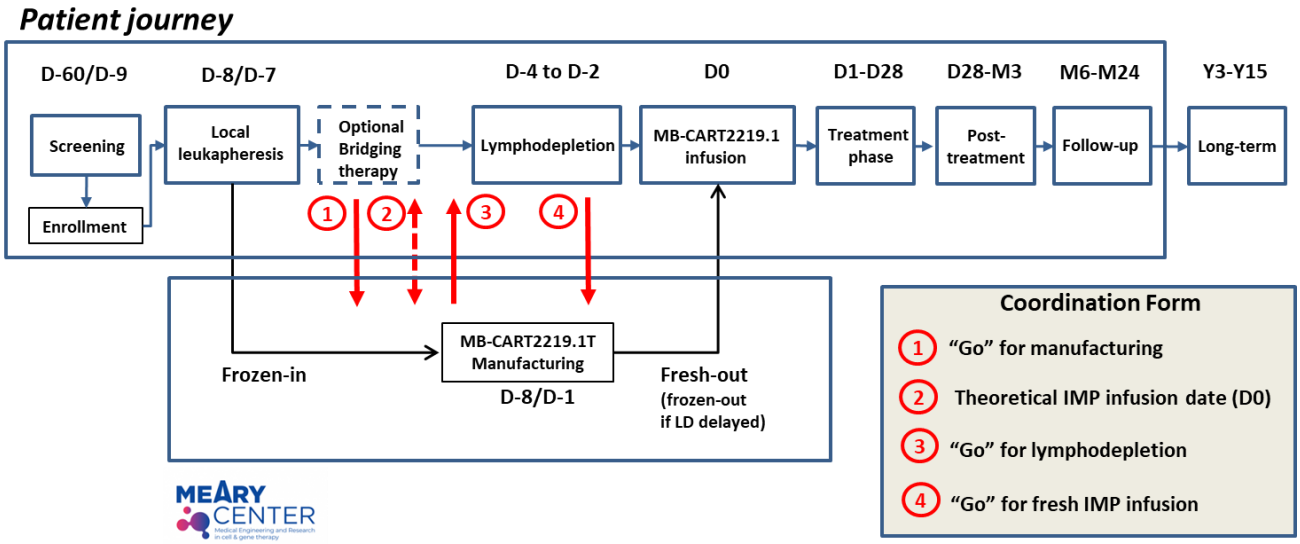
The **lymphodepletion regimen**, administered from **D-4 to D-2**, consists of fludarabine and cyclophosphamide.

Once the IMP is released by the manufacturing platform, it must be infused within **72 hours**.

After the **infusion of MB-CART2219.1** (**D0**), patients undergo frequent monitoring for any signs of adverse events, including TLS, CRS, or ICANS. The monitoring continues in the **post-treatment phase** with detailed **follow-up** visits at regular intervals to assess both the efficacy and safety of the CAR-T therapy.

The long-term follow-up from year 3 to year 15 is not part of this clinical trial but mandatory. It will be submitted as a low intervention clinical trial before the end of the present trial

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Alignment of CAR-T manufacturing and lymphodepletion

Day to infusion	D-8	D-7	D-6	D-5	D-4	D-3	D-2	D-1	D0
CAR-T manuf.	D0	D1	D2	D3	D4	D5	D6	D7	
LD					Flu	Flu/Cy	Flu/Cy		

Manufacturing will start on Monday or thursday

D-8	Manufacturing	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
D-7	*Operator dependant*	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday	Monday
D-4	LD + *Operator dependant*	Friday	Saturday	Sunday	Monday	Tuesday	Wednesday	Thursday
D-1	End of Manufacturing	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
D0	Infusion	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday	Monday

Figure 6. Patient journey and IMP manufacturing.

The following chapters list the different visits of this process.

5.2 Screening Visit (Day -60 to Day -15)

The **screening visit** will take place between **Day -60 and Day -7 prior to the apheresis** procedure Key activities during the screening visit include:

- **Informed Consent:** Patients must provide signed and dated informed consent before any study-specific procedures are performed.
- **Conducted by**: Haematologist and clinical study staff
- **Procedure**: This visit will take place in an outpatient setting or during hospitalization, depending on the patient's clinical condition. During this visit, eligibility assessments will be conducted, including a physical examination, laboratory tests, and bone marrow aspiration.

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Statements about patient's information and consent are described below. The procedure for informing participants and obtaining their consent are described in detail in the section on Ethical and legal considerations.

Whose consent must be obtained	Who informs the individuals and collects their consent	At what point the individuals are informed	At what point the consent is obtained
The subject participating in the trial or legal representative	The investigator or collaborating physician declared and trained in the study (haematologist)	During a standard care visit	At least one day after information

The **screening visit** will occur within **7 days prior to the apheresis** procedure. During this visit, baseline data will be collected to assess the patient's eligibility and establish key characteristics. The following evaluations and tests will be performed:

- Medical history and medication review: Documentation of the patient's medical history, including prior or concomitant medications, chemotherapy regimens, hematopoietic stem cell transplant (HSCT), and graft-versus-host disease (GvHD) history, if applicable.
- Performance status: evaluation of the patient's performance status using the Eastern Cooperative Oncology Group (ECOG) scale.
- Vital signs: measurement of vital signs, including blood pressure, heart rate, respiratory rate, and temperature.
- Physical examination: comprehensive clinical examination to assess the patient's overall health.
- Bone Marrow Aspiration:
    - Karyotype and FISH analysis for genetic abnormalities
    - Immunophenotyping to assess CD19 and CD22 expression
    - Minimal residual disease (MRD) assessment via flow cytometry (FCM) or quantitative PCR (qPCR).
- Imaging: a PET scan is not mandatory and will be conducted if there is suspected or known extramedullary disease.
- Lumbar puncture: assessment of central nervous system involvement.
- Donor chimerism: Assessment of donor chimerism in patients who have previously undergone HSCT, or based on historical results if obtained within the past 3 months.
- Laboratory tests:
    - Complete Blood Count (CBC)
    - Blood Biochemistry: renal function tests (glomerular filtration rate [GFR]), hepatic function tests, lactate dehydrogenase (LDH), uric acid, calcium (Ca), and phosphate (Ph).
    - Coagulation Panel: prothrombin time (PT), activated partial thromboplastin time (aPTT), fibrinogen, and D-dimer levels.
    - Serum Protein Electrophoresis
- Pregnancy test: for women of reproductive potential, a pregnancy test will be performed.
- Infectious disease screening: serology tests for HIV1/2, hepatitis B (HBV), hepatitis C (HCV), cytomegalovirus (CMV), Epstein-Barr virus (EBV), human T-lymphotropic virus 1/2 (HTLV1/2), SARS-Cov2 by PCR on nasal swab and syphilis.
- Cardiac evaluation:
    - Myocardial Function: assessed through ultrasonography or MUGA.
    - Electrocardiogram (ECG)

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- Neurological assessment: if not previously performed, a brain MRI with contrast injection will be completed to evaluate for any existing neurological abnormalities.
- Lymphocyte subset analysis: quantification of T-lymphocyte (CD3+), B-lymphocyte, and natural killer (NK) cell counts.

5.3 Leukapheresis visit (D-10 to D-8)

The leukapheresis visit will take place within 7-9 days before the lymphodepletion regimen and up to 2 days before leukapheresis:

- **Conducted by**: haematologist and apheresis unit specialist
- **Procedure**: the leukapheresis procedure will be carried out in a specialized apheresis unit[3] (hospital or Etablissement Français du Sang (EFS), according to local organizations), typically as an outpatient procedure.

The following evaluations and tests will be performed before apheresis:

- Performance status (ECOG)
- Recent medical history and prior/concomitant medications/chemotherapy
- Physical examination
- Complete blood cell count (CBC) with total lymphocyte count
- Blood biochemistry: renal function with glomerular filtration rate (GFR), hepatic tests (ALP, gammaGT, AST, ALT, bilirubin)
- Coagulation panel: including at least prothrombin time (PT), activated partial thromboplastin time (aPTT), fibrinogen, and D-dimer
- T-lymphocyte (CD3+), B-cell, and NK-cell count

5.4 Pre-lymphodepletion visit (D-7 to D-5)

The pre-lymphodepletion visit will occur within 2 days before starting lymphodepletion regimen:

- **Conducted by**: haematologist and clinical study staff
- **Procedure**: this visit will occur in the hospital and will involve administering the lymphodepletion regimen (fludarabine and cyclophosphamide). The patient will be admitted for hospital monitoring.

The following evaluations and tests will be performed before lymphodepletion:

- Performance status (ECOG)
- Recent medical history and prior/concomitant medications/chemotherapy
- Vital signs
- Physical examination
- Complete blood cell count (CBC)
- Bone marrow aspiration with % blasts and MRD by qPCR or FCM
- Lumbar puncture for CSF cytology assessment
- Blood biochemistry: renal function with GFR, hepatic tests (ALP, gamma GT, AST, ALT, bilirubin), LDH, uric acid, Ca, Ph
- Coagulation panel with at least PT, aPTT, fibrinogen, D-dimer
- Serum protein electrophoresis Immunophenotyping: B/T/NK
- PCR for EBV, CMV, HHV6

[3] The collection of lymphocytes from patients eligible for autologous CAR-T cell therapy may only be performed in an establishment authorized to collect cells for therapeutic purposes, in accordance with Articles R.1242-8 and subsequent articles of the French Public Health Code.

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- Ag tests: Aspergillus, galactomannan
- Pregnancy screen (in females with reproductive potential)
- ECG
- Influenza A & B / SARS-CoV-2 testing (within 10 days prior to infusion, required during the months of October through May)

5.5 Pre-infusion visit (D-2 to D-1)

The pre-infusion visit takes place within 1 day before IMP infusion:

- **Conducted by:** Haematologist
- **Procedure:** the visit will be in the hospital

For rescue therapy following a CAR-T cell infusion, 2 doses of tocilizumab must be readily available.

The following evaluations and tests will be performed before IMP infusion:

- Performance status (ECOG)
- Vital signs
- Physical examination
- Complete blood cell count (CBC)
- Blood biochemistry: renal function with GFR, hepatic tests (ALP, gamma GT, AST, ALT, bilirubin)
- Coagulation panel with at least PT, aPTT, fibrinogen, D-dimer
- Serum protein electrophoresis
- PCR for EBV, CMV, HHV6
- Ag tests: Aspergillus, galactomannan
- Immunophenotyping: B/T/NK
- Immunogenicity against MB-CART2219.1
- Cytokine levels

5.6 Treatment phase visits at D1, D2, D3, D4, D7, D11, D14

The treatment phase visits will be performed from D1 to D4, at D7, D11 and D14 +/- 1 day

- **Conducted by:** Haematologist
- **Procedure:** the visit will be in the hospital

The following evaluations and tests will be performed:

- Performance status (ECOG)
- Vital signs
- Physical examination
- Complete blood cell count (CBC) (with total lymphocyte count)
- Blood biochemistry: renal function with GFR, hepatic tests (ALP, gamma GT, AST, ALT, bilirubin), LDH, uric acid, Ca, Ph
- Coagulation panel with at least PT, aPTT, fibrinogen, D-dimer
- PCR for EBV, CMV, HHV6 (D1, D4, D7, D11, and D14)
- Ag tests: Aspergillus, galactomannan
- Immuno-CART (3x/week depending on the day of infusion)
- Pharmacokinetics of MB-CART2219.1 (qPCR) at D4, D7, D11, D14
- Cytokine levels
- Immunogenicity against MB-CART2219.1 (D14 only)

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5.7 Post-treatment visits at D28, M2 and M3

The post-treatment visits will be performed at D28 +/- 3 days, M2 +/- 7 days, and M3 +/- 7 days):

- **Conducted by:** Haematologist
- **Procedure:** outpatient visit

The following evaluations and tests will be performed:

- Performance status (ECOG)
- Recent medical history and prior/concomitant medications/chemotherapy (M2 and M3)
- Vital signs
- Physical examination
- Bone marrow aspiration: with % blasts, phenotype of the blasts, MRD by qPCR or FCM (at D28 and M3 only)
- PET scan if extramedullary disease (at D28)
- Lumbar puncture for CSF cytology assessment (at D28 only)
- Complete blood cell count (CBC) with differential
- Blood biochemistry: renal function with GFR, hepatic tests (ALP, gamma GT, AST, ALT, bilirubin)
- Coagulation panel with at least PT, aPTT, fibrinogen, D-dimer
- Serum protein electrophoresis
- PCR for EBV, CMV, HHV6
- Ag tests: Aspergillus, galactomannan
- Immuno-CART
- Kinetics of MB-CART2219.1 (qPCR)
- Replication competent lentivirus (RCL, qPCR) at M3 only
- Immunogenicity against MB-CART2219.1 (at D28 and M3 only)

5.8 Follow-up visits including last study visit

Patients will be followed **every 3 months** for **up to 2 years** post-infusion or until disease progression. Afterward, systematic evaluations will be conducted for up to 15 years as recommended by health authorities for patients treated with gene therapies:

- **Conducted by:** Haematologist
- **Procedure:** outpatient visit

The long-term follow-up visits will include the following assessments:

- Performance status (ECOG)
- Recent medical history and prior/concomitant medications/chemotherapy
- Vital signs
- Physical examination
- Bone marrow aspiration: with % blasts, phenotype of the blasts, MRD by qPCR or FCM (at M6 and M12 only)
- Complete blood cell count (CBC) with differential
- Blood biochemistry: renal function with GFR, hepatic tests (ALP, gamma GT, AST, ALT, bilirubin)
- Coagulation panel: including PT, aPTT, fibrinogen, D-dimer
- Serum protein electrophoresis
- Immuno-CART
- PCR for EBV, CMV, HHV6
- Kinetics of MB-CART2219.1 (qPCR)
- Replication competent lentivirus (RCL, qPCR) at M6 only
- Immunogenicity against MB-CART2219.1 (at M6 and M12 only)

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5.9 Expected length of participation and description of the chronology and duration of the clinical trial.

The maximum period between screening and enrolment in the study is 7 days. After enrolment, patients will undergo apheresis to collect T cells for CAR-T manufacturing. Bridging therapy may be administered depending on the patient's disease burden:

- **The treatment phase** begins on the day of the CAR-T cell infusion and lasts for approximately one month.
- **The post-treatment phase** will continue for an additional 2 months, from Day 28 to Month 3.
- **The follow-up period** will take place from Month 3 to Month 24.

In total, patients will be actively followed for 24 months post-infusion, with regular follow-up visits. However, extended monitoring for long-term safety will continue for up to 15 years, in line with regulatory guidelines for gene therapy trials.

Table. Screening, enrolment, participation and study duration

Maximum period between screening and enrolment	7 days
Inclusion period	18 months
Duration of participation for each subject	
• Treatment period	1 month
• Follow-up period	24 months
Total study duration	**42months**
Post trial long-term follow-up (cf. below)	**15 years**

Any patients who will received MB-CART2219.1 during the present clinical trial will have a 15 years follow-up in order to collect safety data. Such follow up will take place as part of low intervention clinical trial. Before the end of the DUALCALM01 clinical trial, the sponsor will request an approval from the ethical committee and an authorization from competent authority on the CTIS platform.

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5.10 Table or diagram summarising the chronology of the study

Actions	Screening	Pre-apheresis	Pre-lymphodepletion	Pre-infusion	Treatment phase				Post-treatment			Follow-up	
				D-2/-1	D1-D4	D7	D11	D14	D28	M2	M3	M6, M12, M24	M9, M15, M18, M21, M24
Information	X*												
Consent form signature	X												
Medical history	X	X	X							X	X	X	X
Performance Status	X	X	X	X	X	X	X	X	X	X	X	X	X
Physical exam	X	X	X	X	X	X	X	X	X	X	X	X	X
Vital signs	X	X	X	X	X	X	X	X	X	X	X	X	X
Adverse events		X	X	X	X	X	X	X	X	X	X	X	X
CBC w/ differential	X	X	X	X	X	X	X	X	X	X	X	X	X
Biochemistry	X	X	X	X	X	X	X	X	X	X	X	X	X
Coagulation test	X	X	X	X	X	X	X	X	X	X	X		
SPE	X			X					X	X	X	X	X
Pregnancy test	X		X										
Serology	X												
PCR CMV, EBV, HHV6	X		X	x		X		X	X	X	X	X	X
LVEF (echo or MUGA)	X												
ECG	X		X										
Brain MRI w/ contrast	X												
BM aspiration	X		X						X		X	X	
BM karyotype + molecular testing	X												
BM MRD	X		X						X		X	X	
CSF assessment	X		X						X				
PET-CT (ALL w/ extramedullary disease)	X								X		X	X	
Chimerism	X												
T-lymphocyte count	X	X											
CD19+ lymphocyte count	X		X			X		X	X	X	X	X	X
Influenza A & B testing			X	X									
Kinetic of CAR-T (qPCR)				X	X	X	X	X	X	X	X	X	X
Cytokine level				X	X	X	X	X					
RCL (qPCR)											X	X	
Immunogenicity against IMP				X				X	X		X	X	

CBC w/ differential: Complete Blood Count with differential; SPE: Serum Protein Electrophoresis; PCR CMV, EBV, HHV6: Polymerase Chain Reaction for Cytomegalovirus, Epstein-Barr Virus, Human Herpesvirus 6; LVEF (echo or MUGA): Left Ventricular Ejection Fraction (via echocardiogram or Multigated Acquisition Scan); ECG: Electrocardiogram; BM: Bone Marrow; MRD: Minimal Residual Disease; CSF: Cerebrospinal Fluid; PET-CT: Positron Emission Tomography–Computed Tomography; RCL: Replication Competent Lentivirus; qPCR: Quantitative Polymerase Chain Reaction.

*Informed consent has to be signed before screening Physical exams including complete neurological evaluation

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5.11 Distinction between standard care and study

5.10.1 Study-Specific Interventions

These interventions are primarily focused on the infusion and evaluation of the investigational dual-targeted CAR-T product (**MB-CART2219.1**) and its safety and efficacy over the long term. The study-specific procedures include:

- **Screening**
- **Leukapheresis**
- **IMP manufacturing**
- **Lymphodepletion**
- **Infusion, treatment and post-treatment phases**
- **Follow-up visits**
- **Research-specific laboratory tests**:
    - Replication-Competent Lentivirus (RCL, qPCR)**:** This test is required to monitor for the presence of replication-competent lentivirus, which is specific to the use of gene-modified products like CAR-T cells.
    - Immunogenicity Testing**:** This test evaluates the potential immune response against the investigational CAR-T product (MB-CART2219.1) and is not a routine part of standard care.
    - Kinetics of CAR-T Cells (qPCR)**:** While general monitoring of CAR-T cell persistence is part of standard care, the detailed analysis of CAR-T kinetics using quantitative PCR (qPCR) is a study-specific measure to assess the behaviour and persistence of the investigational CAR-T cells in the patient

5.10.2 Standard Care Components

The following interventions and procedures are considered standard care for patients receiving CAR-T cell therapy for BCP-ALL:

- Bridging therapy: The period included between apheresis and lymphodepletion, which is let to investigator appreciation.
- Biological tests performed during the different visits excepted those mentioned above (5.10.1)

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Table: "Standard care" vs. "additional interventions" required specifically for the study

Interventions, procedures and treatments carried out for research purposes	**Interventions, procedures and treatments associated with <u>standard care</u>**	**Interventions, procedures and treatments added for <u>research purposes</u>**
Treatments	Bridging therapy	Lymphodepleting regimen Infusion of dual targeted CD19/22 CAR-T cells Treatment phase Follow-up
Visits		Screening, pre-apheresis, pre-lymphodepletion Pre-infusion Treatment phase D28, M2 and M3, M6, M9, M12, M15, M18, M22, M24
Blood tests	All blood tests	
Bone Marrow aspiration	Before apheresis, before lymphodepleting regimen, at D28, M2, M3, M6, M12, M15, M18, M22, M24	
Imaging	PET CT if extramedullar disease at baseline + day 28 Any additional imaging required per investigator decision	Brain MRI at baseline

5.12 Biological samples collection

Samples taken as part of the study will be stored in a biological sample collection.

During the study the sample collection(s) will be stored at CRB, Saint Louis Hospital, Paris, under the supervision of Pr. Emmanuelle Clappier.

At the end of the study, the samples will be kept for a period of 15 years.

At the end of the study, the samples may be used for further analysis not described in the initial protocol but which may be useful for investigation of the condition in light of advances in scientific knowledge, provided the participant is informed and does not oppose this, as stated in the information note/consent form.

If the samples are kept at the end of the study, the sample collection will be declared to the ministry of research and to the director of the competent regional healthcare authority - if the entity is a health establishment] (Article L. 1243-3 *Code de la Santé Publique* [French Public Health Code]).

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Type of sample	Quantity	Storage location (name and entity)	Supervisor of the sample collection (name and entity)	Purpose of the sample collection	Storage duration	End use/Future (e.g. destruction)
Blood, screening	5 mL	CRB, Laboratoire d'Hématologie, Hôpital Saint-Louis, APHP	Pr. Emmanuelle Clappier Laboratoire d'Hématologie , Hôpital Saint-Louis, APHP	Research (CAR-T PK)	15 years	Destruction
Bone Marrow, screening	1 mL					
Blood, pre-LD	5 mL					
Blood, D1-4	5 mL					
Blood, D7	5 mL					
Blood, D11	5 mL					
Blood, D14	5 mL					
Blood, D28	5 mL					

6 ELIGIBILITY CRITERIA

6.1 Inclusion criteria

1. **Age:** Patients must be aged 18 years or older and less than 70 years at the time of inclusion.
2. **Diagnosis:** Patients must have a confirmed diagnosis of relapsed or refractory (R/R) B-cell precursor acute lymphoblastic leukaemia (BCP-ALL) with ≥ 5% bone marrow blasts. The diagnosis should meet one of the following criteria:
    a. Second or greater relapse.
    b. Any relapse after allogeneic stem cell transplantation (SCT), provided the relapse occurs at least 3 months post-SCT.
    c. Refractory at relapse (failure to achieve CR after one cycle of chemotherapy for relapsed leukaemia).

    Patients must have exhausted relevant authorised, on-label and accessible treatment options
3. **Ph+ ALL** must be intolerant or failed two lines of TKI therapy.
4. **CD19 or CD22 Expression:** Patients must demonstrate CD19 expression in bone marrow or peripheral blood by flow cytometry within 3 weeks prior to study inclusion. CD22 expression is not mandatory but should be assessed.
5. **Performance Status:** Patients must have an Eastern Cooperative Oncology Group (ECOG) performance status of 0 to 2.
6. **T-Cell Count:** Patients must have an absolute CD3+ T cell count of ≥100/µL.
7. **Reproductive Status:**
    a. Female patients of childbearing potential must have a negative pregnancy test at screening and before chemotherapy. They must agree to use a highly effective form of birth control from the time of leukapheresis up to 12 months post-MB-CART2219.1 infusion.
    b. Male patients of fathering potential must agree to use a highly effective form of birth control from the time of leukapheresis up to 12 months post-MB-CART2219.1 infusion.
8. **Informed Consent:** Patients must provide signed and dated informed consent prior to the initiation of any trial-specific procedures.

*A woman is considered of childbearing potential (WOCBP), i.e. fertile, following menarche and until becoming post-menopausal unless permanently sterile. Permanent sterilisation methods include hysterectomy, bilateral salpingectomy and bilateral oophorectomy. A postmenopausal

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state is defined as no menses for 12 months without an alternative medical cause. A high follicle stimulating hormone (FSH) level in the postmenopausal range may be used to confirm a post-menopausal state in women not using hormonal contraception or hormonal replacement therapy. However, in the absence of 12 months of amenorrhea, a single FSH measurement is insufficient.

The highly effective methods of contraception authorized for the clinical trial are:

- combined (oestrogen and progestogen containing) hormonal contraception associated with inhibition of ovulation:
  - o oral
  - o intravaginal
  - o transdermal
- progestogen-only hormonal contraception associated with inhibition of ovulation:
  - o oral
  - o injectable
  - o implantable
- intrauterine device (IUD)
- intrauterine hormone-releasing system (IUS)
- bilateral tubal occlusion
- vasectomised partner
- sexual abstinence

6.2 Exclusion criteria

Patients will be excluded from participation in this clinical trial if they meet any of the following criteria:

1. **CNS Involvement:** Disease involvement of the central nervous system at the time of MB-CART2219.1 lymphodepletion.
2. **Recent Medication:** One of the following previous medications:
   - Systemic chemotherapies, corticosteroids (with the exception of physiologic replacement dosing), tyrosine kinase inhibitors (TKI), intrathecal therapy within 7 days prior to leukapheresis
   - Fludarabine/clofarabine or immunosuppressive drugs (e.g., calcineurin inhibitors) and antibodies or antibody-targeted agents (e.g., rituximab, daratumomab, isatuximab, blinatumomab, inotuzumab-ozogamicin) or donor lymphocyte transfusions or radiation therapy within 30 days prior to leukapheresis
   - Live vaccines within 30 days prior to leukapheresis
   - Alemtuzumab within 3 months prior to leukapheresis,
3. Systemic chemotherapies or corticosteroids (with the exception of physiologic replacement dosing) or tyrosine kinase inhibitors (TKI) or intrathecal therapy within 7 days prior to leukapheresis.
4. Fludarabine/clofarabine or immunosuppressive drugs (e.g., calcineurin inhibitors) and antibodies or antibody-targeted agents (e.g., rituximab, daratumumab, isatuximab, blinatumomab, inotuzumab ozogamicin) or donor lymphocyte transfusions or radiation therapy within 30 days prior to leukapheresis.
5. Live vaccines within 30 days prior to leukapheresis.
6. Alemtuzumab within 3 months prior to leukapheresis.

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7. **Other Investigational Product:** Administration of other investigational products between apheresis and MB-CART2219.1 infusion.
8. **Active CNS Disease:** Active clinically significant CNS disease (including, but not limited to, uncontrolled seizure disorders, cerebrovascular ischemia or hemorrhage, dementia, paralysis).
9. **Active GVHD or Immunosuppression:** Active graft-versus-host disease (GVHD) or immunosuppression requirement (hydrocortisone > 12 mg/m²/day or equivalent).
10. **Organ Failure:**
    - Cardiac ejection fraction < 45%.
    - Calculated glomerular filtration rate < 30 mL/min.
    - Bilirubin > 3 times the upper limit of normal.
11. **Uncontrolled Cancer:** History of uncontrolled cancer within the last 2 years, with the exception of basal cell carcinoma or carcinoma in situ of the cervix.
12. **Viral infection:** Known positive serology for HIV-1 and -2, HTLV-1 and -2, or active hepatitis B or C infection.
13. Positive SARS-Cov2 by PCR on nasal swab,
14. **Autoimmune Disease:** Personal history of autoimmune disease.
15. **Contraindications:** Contraindication of investigational medicinal product, of auxiliary medicinal products and of trial-related procedures.
16. **Reproductive Potential:**
    - Female patients of childbearing potential not willing to practice a highly effective form of birth control from leukapheresis up to 12 months after MB-CART2219.1 infusion.
    - Male patients of fathering potential not willing to practice a highly effective form of birth control from leukapheresis up to 12 months after MB-CART2219.1 infusion.
17. **Concurrent Clinical Trials:** Concurrent participation in another interventional trial that could interact with this trial (e.g., CAR-T trials).

6.3 Recruitment procedure

Recruitment Strategy:

- **Participating Centres:** The clinical trial will be conducted across five clinical sites in France, each with a robust infrastructure and extensive experience in conducting clinical trials in haematology and advanced therapy medicinal products (ATMPs):
    - Paris Saint-Louis Hospital
    - Lyon
    - Bordeaux
    - Marseille
    - Nantes
- **Referral networks:** Each participating site is integrated into national and regional referral networks, including cooperative groups such as GRAALL (Group for Research on Adult Acute Lymphoblastic Leukaemia). These networks will facilitate the identification and referral of eligible patients.
- **Eligibility screening:** Potential participants will be screened based on predefined inclusion and exclusion criteria. Screening will be conducted during routine clinical visits or specific screening appointments.

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Recruitment justification:

- **Clinical sites and expertise:** Each of the five clinical sites is a leading centre in haematology and has demonstrated significant recruitment capabilities in previous clinical trials. Their combined experience ensures that they can efficiently recruit, manage, and monitor the participants.
- **Patient population:** The clinical trial targets adult patients with relapsed/refractory (R/R) B-cell precursor acute lymphoblastic leukaemia (BCP-ALL), a patient population with a substantial unmet medical need. Given the high incidence of relapse in this patient group and the lack of effective treatment options, there is a strong motivation for patients and physicians to participate in innovative clinical trials such as this one.
- **Recruitment projections:** Each site is expected to recruit approximately 4 patients over the course of the study. With five sites participating, the study aims to recruit a total of 20 patients. The recruitment rate is projected at 0.22 patients per site per month, which is feasible based on historical data from previous trials and the patient load at these centres.

	Number of participants
Total number of participants to be included	20
Number of centres	5
Enrolment period (months)	18
Number of participants/centre	4
Number of participants/centre/month	0.22

- **Patient engagement:** Comprehensive patient engagement strategies, including educational materials and consultations, will be employed to inform patients about the trial. This approach aims to enhance patient understanding and willingness to participate.

6.4 Termination rules

6.4.1 Termination rules for individual participants

6.4.1.1 Criteria and procedures for temporary suspension of the study treatment and continuation of the participation from the study

Temporary suspension of the treatment is allowed at all stages of the treatment, especially to recover from adverse events.
The treatment can be resumed once the adverse event has been resolved.

If, before CAR-T infusion, the participant presents an adverse event that preclude lymphodepletion or CAR-T infusion, the study procedure must be temporary discontinued but the participant will continue to be monitored for the study.

The investigator must:

- Document the reason(s)
- Collect the assessment criteria at the time of temporary suspension of the study treatment
- Schedule a follow-up for the participant, particularly in case of a serious adverse event.

In a general point of view, dose adaptation or the non-administration of treatment due to the occurrence of toxicity does not lead to premature discontinuation of the study.

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In the case of serious adverse events, the investigator must notify the sponsor and follow up the participant until the severe adverse events are resolved. Notification of a serious adverse event must be sent by email (eig-vigilance.drc@aphp.fr) to the sponsor.

6.4.1.2 Criteria and procedure for definitive withdrawal of the study treatment and continuation of the participation from the study

If, before top start lymphodepletion or before CAR-T infusion, the participant presents one of the following adverse events, then the study product/procedure must be definitively discontinued but the participant will continue to be monitored for the study except in the situations described below (see section 6.4.1.3).

- Active, uncontrolled infection,
- Clinically significant central nervous system (CNS) involvement,
- Severe organ dysfunction, including:
  - Cardiac: left ventricular ejection fraction < 45% or clinically significant cardiac event (e.g., heart failure, arrhythmia)
  - Renal: acute kidney injury or severe renal impairment (eGFR < 30 mL/min/1.73 m^2)
  - Hepatic: bilirubin > 3 × ULN or severe hepatic impairment
- Severe deterioration of general condition:
  - ECOG performance status > 2
  - Rapid clinical deterioration incompatible with study procedures
- Uncontrolled disease progression making study treatment inappropriate, as judged by the investigator,
- Pregnancy,
- Any other serious medical condition that, in the investigator's judgment, would compromise patient safety or compliance with the study procedures

In case of serious adverse events, the investigator must notify the sponsor and follow up the participant until the serious adverse events are resolved.

Participant lost to follow-up: the participant cannot be located. The investigator must make every effort to reconnect with the participant (and document his attempts in the source file), at least to determine whether the participant is alive or dead (for example: to get in contact with the birth city hall).

6.4.1.3 Criteria and procedure for exit the study

- Participants may exit the study at any time and for any reason.
- The investigator can permanently withdraw a participant from the study for any safety reason or if it is in the participant's best interests.
- If a participant exits the study prematurely and whether or not to withdraw consent, any data collected prior to the date of premature exit may still be used.
- If a participant exits the study prematurely, withdraws consent and objects to the use of his/her data, any safety data collected prior to the date of premature exit may still be used.
- In case of serious adverse events, the investigator must notify the sponsor and follow up the participant until the severe adverse events are resolved.

The case report form must list the various reasons why the participant has discontinued the study:

- Adverse reaction (before lymphodepletion or CAR-T infusion)
- Another medical issue

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- Personal reasons of the participant
- Explicit withdrawal of consent
- Lost to follow-up

If a participant discontinues treatment or exits the study, their standard medical care for their condition will continue without interruption or prejudice. The withdrawal from the study will not impact the level or quality of care they would normally receive for their underlying illness.

In the event of SAEs following the premature discontinuation of treatment or participation, the investigator is responsible for continuing to monitor the participant according to the guidelines outlined in Section 6.4.1. This includes documenting and reporting the event to ensure patient safety and regulatory compliance. The monitoring of any unresolved adverse events will continue until the event is resolved or deemed stable.

If a participant is lost to follow-up, the investigator must make every effort to reconnect with the participant (and document his attempts in the source file), at least to determine whether the participant is alive or dead (for example: to get in contact with the birth city hall).

6.4.2 Termination rules for the clinical trial: full or partial discontinuation of the clinical trial

AP-HP as sponsor or the Competent Authority can prematurely discontinue all or part of the study, temporarily or permanently, further to the recommendations of the Data Safety Monitoring Board if suspected unexpected serious adverse reactions (SUSARs) are observed requiring a reassessment of the benefit-risk ratio for the study,

Similarly, AP-HP, as the sponsor, or the Competent Authority may decide to prematurely discontinue the study due to unforeseen issues or new information about the product, in light of which the objectives of the study are unlikely to be achieved.

AP-HP as sponsor reserves the right to permanently suspend enrolment at any time if it appears that the inclusion objectives are not met.

If the study is prematurely discontinued, the decision and justification will be provided by the sponsor (AP-HP) to the Competent Authority and to the Ethics Committee in CTIS platform, without undue delay but not later than in 15 days of the date of the temporary halt or early termination.
If the study is prematurely discontinued for safety reasons, it shall include the reasons for such action and specify follow-up measures, along with recommendations from the Data Safety Monitoring Board in the case of substantial modification.

In case of a protocol substantial modification is necessary, it should be shared to the Competent Authority and to the Ethics Committee in CTIS platform, without undue delay but not later than 15 days of the date of the temporary halt or early termination.

In case of occurrence of an unexpected event that could have a serious impact on the risk/benefit of the trial, this unexpected event should be shared without undue delay but not later than 15 days from the date the sponsor became aware of this event.

In case of implementation of urgent safety measures, they should be taken immediately and should be shared to the Competent Authority and to the Ethics Committee in CTIS platform, without undue delay but no later than 7 days of the date of the measure's implementation.

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7 TREATMENT(S) ADMINISTERED TO STUDY PARTICIPANTS

7.1 Bridging therapy and lympodepletion

1. Bridging Chemotherapy (optional): From leukapheresis to D-9

Bridging chemotherapy may be administered to patients with a high disease burden at screening, defined by M3 marrow (with >25% leukemic blasts) and/or ≥10 G/L blasts in peripheral blood. The goal of this optional bridging therapy is to reduce tumour burden, prevent critical TLS, and minimize the risk of severe CRS or ICANS following CAR-T cell infusion.

- **Discussion and customization:** the decision to administer bridging therapy will be made on a case-by-case basis, discussed collegially by the treating physician and steering committee. The choice of chemotherapy will be based on the patient's medical history and the specific characteristics of ALL, including oncogenetics, immunophenotype, and any extramedullary involvement.
- **Re-evaluation before lymphodepletion:** at the end of the planned bridging chemotherapy, the patient's disease status will be re-evaluated by bone marrow aspiration. Lymphodepletion will only begin once the patient is considered clinically stable and ready to proceed with CAR-T cell therapy.
- **Coordination with manufacturing:** after re-evaluating the patient's readiness for lymphodepletion, the manufacturing site will be informed to launch the final manufacturing steps for the MB-CART2219.1 CAR-T cells. Local pharmacists will coordinate with the clinical team to plan the infusion **(see 5.1, Coordination Form)**.
- **Lymphodepletion:** lymphodepletion will start on Day -4 (D4 of manufacturing) after confirmation that the manufacturing started **(see 5.1, Coordination Form)**.

2. Lymphodepletion (LD): D-4 to D-2

All patients will undergo a lymphodepleting (LD) regimen prior to the infusion of MB-CART2219.1 to induce lymphopenia and promote the engraftment and expansion of the infused CAR-T cells. The combination of fludarabine and cyclophosphamide has been selected based on its established effectiveness in facilitating adoptive immunotherapy, including CAR-T cell therapies.

The lymphodepleting regimen will begin on Day -4, after successful in-process control (IPC) of CAR-T cell transduction and will include the following:

- **Fludarabine:** 40 mg/m²/day intravenously (IV) on Days -4, -3, and -2.
- **Cyclophosphamide:** 500 mg/m²/day IV on Days -3 and -2.

A minimum interval of 48 hours between the last cyclophosphamide dose and the CAR-T infusion must be observed.

Alignment of CAR-T manufacturing and lymphodepletion

Day to infusion	D-8	D-7	D-6	D-5	D-4	D-3	D-2	D-1	D0
CAR-T manuf.	**D0**	**D1**	**D2**	**D3**	**D4**	**D5**	**D6**	**D7**	
LD					Flu	Flu/Cy	Flu/Cy		

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Dose adjustments

- For patients with creatinine clearance between 30 and 70 mL/min/1.73 m², the fludarabine dose will be reduced by up to 50%. Close monitoring of creatinine clearance and renal function will be performed daily, according to institutional standards.

Alternative timing for lymphodepletion

- If the lymphodepletion regimen cannot be initiated on Day 4 of IMP manufacturing, the IMP will be cryopreserved and infused at a later time after the lymphodepletion has been successfully completed.

Chemotherapy agents

- **Fludarabine:** Fludarabine phosphate is a synthetic purine nucleoside analog that functions as an antimetabolite. Its sugar moiety is arabinose, differentiating it from physiologic nucleosides. Fludarabine is used to inhibit DNA synthesis, inducing lymphopenia and aiding CAR-T cell engraftment.
- **Cyclophosphamide:** Cyclophosphamide is an alkylating agent derived from nitrogen mustard. It requires activation by the liver into active metabolites, which then function to alkylate DNA and induce immunosuppression. Cyclophosphamide is frequently used in lymphodepleting regimens due to its ability to suppress the immune system and enhance adoptive cell therapies. The serum half-life ranges from 3 to 12 hours, with metabolites detectable in the serum up to 72 hours post-administration.

Fludarabine and cyclophosphamide will be provided by the clinical sites and labelled as per their local procedures.

Refer to the current product label for guidance on packaging, storage, preparation, administration and toxicity management associated with the administration of chemotherapy agents.

7.2 Investigational medical product : MB-CART2219.1

7.2.1 Nature of the experimental ATMP

The IMP, **MB-CART2219.1**, is a **dual-targeted CAR-T cell therapy** designed to target both CD19 and CD22 antigens on B-cell malignancies. This **experimental ATMP** is derived from the patient's own (autologous) T cells, which are genetically modified ex vivo using a lentiviral vector to express a chimeric antigen receptor (CAR) specific for both CD19 and CD22. It is categorized as gene therapy medicinal product (GTMP).

7.2.2 How the ATMP is obtained

1. **Leukapheresis**: T cells are collected from the patient through a **leukapheresis** procedure. A minimum of $50x10^6$/mm3 CD4+CD8+ is needed to start the manufacturing. A second day or a third day of leukapheresis can be done if needed to obtain the minimal cell count.
2. **Genetic Modification and Expansion**: The collected T cells are then sent to a centralized manufacturing facility (**Paris Saint-Louis Hospital ATMP Platform**) where they are lentivirally transduced to express the dual-targeted CAR. The genetically modified cells are expanded and prepared for infusion.
3. **Final Product**: The resulting **MB-CART2219.1** CAR-T cells are formulated into either **fresh or cryopreserved** batches, depending on the timing of lymphodepletion and the patient's readiness for infusion **(see 5.1, Coordination Form)**.

7.2.3 Supply form and characteristics

- **Form**: The **MB-CART2219.1** product is supplied in **600mL bag** for **fresh infusion or in a cryostorage bag for cryopreserved product**.

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- If **fresh**: The cells are infused directly into the patient upon completion of manufacturing **(see 5.1, Coordination Form)**.
- If **cryopreserved**: The product is frozen and stored until the patient is ready for infusion.

- **Volume**: The final volume is 150 mL of cell suspension for fresh administration and 20 mL of cell suspension for the cryopreserved product.

7.2.4 Finished product characteristics

- **Posology (Dosage)**: The dose of **MB-CART2219.1** is administered in a single IV infusion, with the dose adjusted based on the patient's body weight:
  - **Dose Level 0**: 0.25 x 10^6 CAR-T cells/kg
  - **Dose Level 1**: 0.5 x 10^6 CAR-T cells/kg
  - **Dose Level 2**: 1.0 x 10^6 CAR-T cells/kg
  - **Dose Level 3**: 2.0 x 10^6 CAR-T cells/kg
- **Expected quantity**: The final product is expected to deliver a minimum of **0.25 x 10^6 CAR-T cells/kg** and a maximum of **2.0 x 10^6 CAR-T cells/kg** depending on the dose level assigned to the patient. The final cell concentration in the bag is adjusted to ensure the proper dosing based on patient weight.
- **Concentration**: The concentration of CAR-T cells varies depending on the efficiency of the manufacturing process, with a typical range of **0.25 to 2.0 x 10^6 cells/kg** of body weight per infusion.
- **Stability**:
  - **Fresh product** must be administered as soon as possible after preparation.
  - **Cryopreserved product** can be stored in liquid nitrogen until the patient is ready for the lymphodepleting chemotherapy and CAR-T cell infusion.

7.3 Description of auxiliary medicinal products

Fludarabine and cyclophosphamide, used for lymphodepletion and already described above, are also considered as auxiliary medicinal product.

7.3.1 Tumour lysis syndrome (TSL) prophylaxis and treatment

TLS is a potentially life-threatening complication that can occur during the lymphodepletion phase and post-infusion of CAR-T cells, particularly in patients with a high tumour burden. To mitigate this risk, careful monitoring and prophylactic interventions are necessary.

Screening Phase:

- **Prophylaxis**:
  - **Allopurinol** or an alternative non-allopurinol agent (e.g., **febuxostat**) should be administered to patients with elevated uric acid levels or high tumour burden to prevent hyperuricemia.
  - **Increased oral and/or intravenous (IV) hydration** should be provided prior to lymphodepleting chemotherapy and CAR-T infusion to facilitate renal clearance and reduce the risk of TLS.
- **Management of early symptoms**:

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- In cases of rising uric acid levels or early symptoms of acute TLS, prompt supportive care is crucial. IV hydration should be intensified, and rasburicase may be administered if uric acid continues to rise despite allopurinol/febuxostat and fluids.
- Early intervention with rasburicase is recommended in patients with high tumour burden or elevated baseline uric acid.

Post-infusion monitoring phase:

- Laboratory monitoring**:** Frequent monitoring of biochemical parameters is essential to detect early signs of TLS. Standard TLS monitoring lab tests should be performed **2-3 times per week for 3 weeks** after the start of lymphodepleting chemotherapy, followed by **weekly monitoring**:
- Hydration**:** adequate hydration should be maintained post-infusion to help reduce the risk of TLS.

7.3.2 Cytokine release syndrome (CRS) and treatment

CRS is a systemic inflammatory response that occurs due to the activation and expansion of CAR-T cells and the subsequent release of cytokines such as IL-6, TNF, and IFN-γ. CRS is a well-known side effect of CAR-T cell therapy and can range from mild symptoms to life-threatening conditions. Effective diagnosis and timely intervention are crucial for managing CRS.

Diagnosis of CRS:

The diagnosis of CRS is based on clinical symptoms and laboratory results. The following clinical manifestations are indicative of CRS:

- Fever**:** CRS often begins with a high fever (≥38°C), which is the hallmark symptom.
- Cardiovascular**:** Symptoms such as hypotension, tachycardia, or widened pulse pressure, which may progress to requiring vasopressors.
- Respiratory**:** Symptoms such as tachypnea or hypoxia, ranging from mild dyspnea to the need for high-flow oxygen or mechanical ventilation.
- Gastrointestinal**:** Nausea, vomiting, diarrhea.
- Coagulation abnormalities**:** Elevated D-dimer, hypofibrinogenemia, or clinical signs of coagulopathy.
- Renal dysfunction
- Hepatic Dysfunction**:** Elevated liver enzymes (transaminitis), hyperbilirubinemia.
- End organ dysfunction**:** Severe cases of CRS may lead to multi-organ failure, requiring intensive care.

Before diagnosing CRS, other potential causes of fever, hypotension, and respiratory distress, such as infections should be excluded.

Monitoring for CRS:

Participants should be closely monitored after CAR-T cell infusion for signs and symptoms of CRS, with special attention to early indicators such as fever and hypotension. Continuous assessment of vital signs and regular blood tests (CBC, biochemistry, coagulation, cytokine levels) should be conducted. Monitoring of cardiac function is also recommended for participants with prolonged or severe CRS.

Grading of CRS (Based on ASTCT Consensus)[29]:

Parameter	Grade 1	Grade 2	Grade 3	Grade 4
Fever[1]	Temperature ≥38°C	Temperature ≥38°C	Temperature ≥38°C	Temperature ≥38°C

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With Hypotension	None	Not requiring vasopressors	Requiring a vasopressor with or without vasopressin	Requiring multiple vasopressors (excluding vasopressin)
and/or[2] Hypoxia	None	Requiring low-flow nasal cannula[3] or blow-by	Requiring high-flow nasal cannula, facemask, non-rebreather mask, or Venturi mask	Requiring positive pressure (e.g., CPAP, BiPAP, intubation, mechanical ventilation)

Organ toxicities associated with CRS may be graded according to CTCAE v5.0 but they do not influence CRS grading.

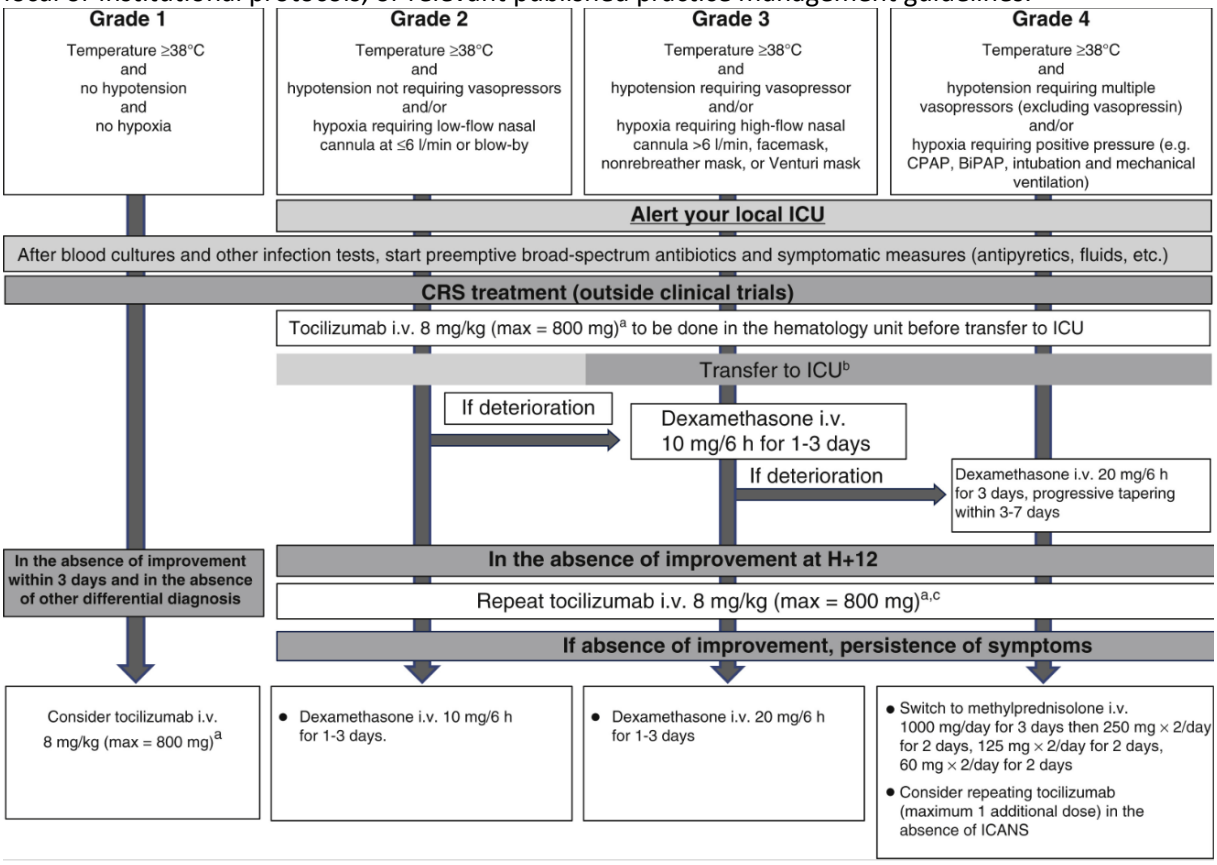
1 Fever is defined as temperature ≥38°C not attributable to any other cause. In participants who have CRS then receive antipyretic or anti-cytokine therapy such as tocilizumab or steroids, fever is no longer required to grade subsequent CRS severity. In this case, CRS grading is driven by hypotension and/or hypoxia.

2. CRS grade is determined by the more severe event: hypotension or hypoxia not attributable to any other cause. For example, a participant with temperature of 39.5°C, hypotension requiring 1 vasopressor, and hypoxia requiring low-flow nasal cannula is classified as grade 3 CRS.

3. Low-flow nasal cannula is defined as oxygen delivered at _6 L/minute. Low flow also includes blow-by oxygen delivery, sometimes used in paediatrics. High-flow nasal cannula is defined as oxygen delivered at >6 L/minute.

Treatment of CRS:

The CRS management algorithm published by Hayden et al.[30] serves as a guideline, and the investigator may exercise discretion or adjust the treatment approach as necessary for an individual patient. Additionally, alternative CRS management strategies may be employed based on clinical experience, local or institutional protocols, or relevant published practice management guidelines.



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Additional Considerations:

- Concurrent neurotoxicity**:** if neurotoxicity (e.g., seizures, confusion) occurs alongside CRS, it should be treated as a separate event. Neurotoxicity management does not influence CRS grading or resolution.

7.3.3 Immune Effector Cell-Associated Neurotoxicity Syndrome (ICANS)

Immune Effector Cell-Associated Neurotoxicity Syndrome (ICANS) is a known complication following MB-CART2219.1 therapy. Neurotoxic events can occur alongside or independent of CRS and may develop during the early stages or after CRS resolution, typically within 4 weeks of infusion. These neurotoxic events are diverse in presentation, ranging from mild confusion to more severe manifestations such as seizures or coma.

Pathophysiology of ICANS:

The exact pathophysiology of ICANS remains poorly understood, but it is believed to result from a supraphysiologic immune response following CAR-T cell therapy. High levels of circulating cytokines, such as IL-6 and IFN-γ, lead to endothelial activation and disruption of the blood-brain barrier, triggering an inflammatory response within the central nervous system (CNS). This cascade results in the clinical manifestations of ICANS, which can vary from mild to severe neurologic dysfunction.

Clinical Manifestations of ICANS:

ICANS presents with a broad spectrum of neurologic symptoms, which may initially include:

- Inattention, disorientation, and mild aphasia (expressive or receptive).
- Progressive symptoms can lead to lethargy, severe aphasia, seizures, and in severe cases, coma.
- The most severe cases may necessitate intubation and mechanical ventilation for airway protection due to impaired consciousness.
- Although the majority of ICANS cases are reversible, uncontrolled neurotoxicity may result in cerebral oedema, brain herniation, and, in rare cases, death.

Diagnosis and Evaluation of ICANS:

The evaluation of participants presenting with ICANS includes:

- Laboratory testing**:** Elevated levels of serum ferritin**,** C-reactive protein**,** and pro-inflammatory cytokines such as IL-1, IL-6, and TNFα.
- Cerebrospinal Fluid (CSF)**:** Increased protein and cell counts, although these findings are usually non-specific.
- Electroencephalogram (EEG)**:** may show frontal or diffuse theta-delta slowing**,** indicative of neurotoxicity but non-specific.
- Neuroimaging**:** CT **or** MRI may reveal diffuse white matter changes **or** sulcal effacement**,** suggestive of cerebral oedema. Imaging may suggest differential diagnoses.

ICANS Grading[4]:

ICANS is graded according to the American Society for Transplantation and Cellular Therapy (ASTCT) guidelines, using the ICE score (Immune Effector Cell-Associated Encephalopathy score), which is a 10-point assessment that evaluates neurologic status.

[4] [4] Lee D., Santomasso B.D., Locke F.L., et al. (2019) ASTCT Consensus Grading for Cytokine Release Syndrome and Neurologic Toxicity Associated with Immune Effector Cells. Biol Blood Marrow Transplant 25, 625-38.

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Domain	Assessment Criteria	Score
Orientation	Ability to orient to year, month, city, hospital	4 points
Naming	Ability to name 3 objects (e.g., clock, pen, button)	3 points
Following Commands	Ability to follow simple commands (e.g., "show me 2 fingers" or "close your eyes and stick out your tongue")	1 point
Writing	Ability to write a standard sentence (e.g., "Our national bird is the bald eagle")	1 point
Attention	Ability to count backwards from 100 by 10	1 point

ICANS is assessed across five domains:

Domain	Grade 1	Grade 2	Grade 3	Grade 4
ICE Score	7 to 9	3 to 6	0 to 2	0 (participant is unarousable and unable to perform ICE)
Depressed level of consciousness	Awakens spontaneously	Awakens to voice	Awakens only to tactile stimulus	Participant is unarousable or requires vigorous or repetitive tactile stimuli to arouse; stupor or coma
Seizure	N/A	N/A	Any clinical seizure (focal or generalized) that resolves rapidly; nonconvulsive seizures on EEG that resolve with intervention	Life-threatening prolonged seizure (>5 minutes), or repetitive clinical or electrical seizures without return to baseline in between
Motor findings	N/A	N/A	N/A	Deep focal motor weakness such as hemiparesis or paraparesis
Elevated ICP/cerebral edema	N/A	N/A	Focal/local edema on neuroimaging	Diffuse cerebral edema on neuroimaging, decerebrate or decorticate posturing, cranial nerve VI palsy, papilledema, or Cushing triad

Management of ICANS:

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Most cases of ICANS are transient and can be managed with supportive care. In more severe cases, the following protocol should be followed:

- **Dexamethasone** should be administered in cases of:
  - **Grade 3 ICANS** (other than headache) persisting for more than 24 hours.
  - **Grade 4 ICANS** of any duration.
  - **Generalized seizures**.
  - The recommended dose **is** 6 mg/m² every 8 hours, with a maximum of 10 mg every 8 hours**.**

Neurologic toxicity occurring concurrently with CRS should be managed in accordance with CRS management guidelines. When CRS has resolved, ICANS should be monitored and treated based on clinical findings and graded according to severity. Continuous monitoring of neurologic function using the ICE score and frequent assessments such as EEG**,** MRI**,** or CT scan are recommended to ensure appropriate management.

7.3.4 Immune effector cell-associated hemophagocytic lympho-histiocytosis-like syndrome (IEC-HS)

Definition:
IEC-HS, also referred to as macrophage activation syndrome (MAS), is a rare but life-threatening hyperinflammatory complication that may occur after CAR-T cell infusion. It is characterized by uncontrolled immune activation, excessive cytokine release, and multi-organ involvement. HLH can overlap with severe CRS and must be promptly recognized and treated due to its high mortality.

Diagnostic criteria:
Diagnosis will rely on adapted HLH-2004 criteria and ASTCT recommendations for IEC-HS, including:

- Persistent fever > 38.5°C not explained by infection,
- Cytopenias affecting ≥ 2 lineages (Hb < 90 g/L, platelets < 100 G/L, neutrophils < 1 G/L),
- Hyperferritinemia (ferritin > 10,000 µg/L, often rapidly rising),
- Hypofibrinogenemia and/or hypertriglyceridemia,
- Elevated soluble IL-2 receptor (sCD25) and CXCL9 when available,
- Evidence of hemophagocytosis on bone marrow aspirate or tissue biopsy (not mandatory for diagnosis in the CAR-T setting).

Monitoring strategy:
Systematic monitoring of ferritin, triglycerides, fibrinogen, and D-dimers at baseline, post-infusion, and in case of fever or cytopenia. Daily monitoring during CRS or suspected HLH episodes is recommended.

IEC-HS grading:

	Grade 1	Grade 2	Grade 3	Grade 4	Grade 5
IEC-HS*	Asymptomatic or mild symptoms; requires observation and/or clinical and diagnostic evaluations. Intervention not indicated.	Mild to moderate symptoms, with intervention indicated. (e.g., immunosuppressive agents directed at IEC-HS, transfusions for asymptomatic hypofibrinogenemia)	Severe or medically significant, but not immediately life threatening (e.g., coagulopathy with bleeding requiring transfusion support, or hospitalization required for new onset acute kidney injury, hypotension or respiratory distress)	Life-threatening consequences: urgent intervention indicated (e.g., life-threatening bleeding or hypotension, respiratory distress requiring intubation, dialysis indicated for acute kidney injury)	Death

*Not attributable to other causes; Defined by the development of pathological and biochemical features of macrophage activation/HLH that is attributable to IEC therapy and associated with progression or new onset of cytopenia,

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hyperferritinemia, coagulopathy with hypofibrinogenemia and hepatic transaminitis (>5 x ULN)). While HLH-like manifestations are frequently seen in patients with severe CRS (as defined by ASTCT), IEC-HS is often delayed in onset and manifests as CRS is resolved/resolving

Treatment algorithm:

- First-line: High-dose corticosteroids (dexamethasone 10 mg/m²/day IV, tapered according to response).
- Second-line / steroid-refractory cases:
    - Anakinra (IL-1 receptor antagonist) as preferred cytokine blockade,
    - Ruxolitinib or etoposide may be considered in refractory situations after multidisciplinary discussion.

Supportive care: broad-spectrum antibiotics (to rule out infection), transfusion support, management of organ dysfunction in ICU. Tocilizumab is usually not effective in HLH without concomitant CRS.

Distinction from CRS/ICANS:
Although HLH can overlap with CRS, the disproportionate elevation of ferritin, profound cytopenias, and coagulopathy help differentiate it. Prompt differentiation is critical, as HLH often requires early corticosteroids and cytokine blockade rather than IL-6 inhibition alone.

Reporting:
All suspected HLH/MAS events will be considered Adverse Events of Special Interest (AESI) and must be reported to the sponsor within 24 hours.

7.3.5 Hypogammaglobulinemia

Due to the nature of the therapy targeting both malignant and non-malignant B-cells, B-cell aplasia is a common and expected outcome in all responding patients. The loss of B-cells serves as a useful surrogate for CAR-T cell persistence and treatment efficacy but can result in hypogammaglobulinemia or agammaglobulinemia, increasing susceptibility to infections, particularly from encapsulated organisms and viral reactivations such as herpes viruses.

The prolonged depletion of normal B-cells, especially in patients with durable responses to MB-CART2219.1, can persist for an extended period, reflecting the long-term persistence of memory CAR-T cells. Given the decade-long lifespan of memory T-lymphocytes, some patients may experience sustained B-cell aplasia, with long-term risks of infection.

Management of hypogammaglobulinemia includes:

- Monitoring immunoglobulin levels regularly post-infusion, particularly beyond Day 28.
- Infection prevention measures, including antibiotic prophylaxis to protect against bacterial infections, and vigilant monitoring for viral reactivation.
- For patients with persistent B-cell aplasia and hypogammaglobulinemia (< 4 g/L), intravenous immunoglobulin (IVIg) replacement therapy will be administered to reduce the risk of infection.

This strategy ensures patients at risk of prolonged B-cell depletion and hypogammaglobulinemia are protected from infection while maintaining the therapeutic benefits of MB-CART2219.1.

7.4 Other supportive cares

Transfusion Support:
Blood product support, including red blood cell (RBC) and platelet transfusions, may be administered based on institutional standards to manage anaemia or thrombocytopenia. These transfusions will be considered concomitant procedures and should be documented accordingly in the participant's eCRF (electronic Case Report Form).

Plasma Fresh Frozen (PFC) may also be administered as needed in cases of coagulopathy to manage clotting disorders.

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Anti-Infectious Therapy:
Preventive and curative anti-infectious therapy, including antibiotics, antivirals, and antifungals, should be administered following the European Conference on Infections in Leukaemia (ECIL) recommendations. The latest guidelines can be accessed on the ECIL website: www.kobe.fr/ecil.

Recommendations:

- Prophylaxis:
  - Valaciclovir and trimethoprim-sulfamethoxazole (or equivalent) must be continued or initiated due to lymphopenia, in accordance with local protocols.
- Antifungal Therapy:
  - Prophylactic antifungal therapy with posaconazole should be initiated at a dose of 300 mg twice daily (oral or IV) on the first day, followed by 300 mg once daily thereafter. This should be continued from Day 0 until neutrophil recovery.
- G-CSF (Granulocyte Colony-Stimulating Factor):
  - G-CSF at 5 µg/kg/day should be administered if the absolute neutrophil count (ANC) is below 500/mm³ on Day 14, continuing until the ANC is greater than 1000/mm³ for three consecutive days.

Antiemetic Medication:
Prophylactic antiemetic medication should be administered prior to the start of the conditioning regimen to prevent nausea and vomiting, following institutional standards. This may include a 5-HT3 receptor antagonist or other locally available and appropriate antiemetic agents. Antiemetics will be considered concomitant treatments and should be recorded in the participant's eCRF.

7.5 Prohibited treatments

7.5.1 Prohibited treatments prior to apheresis

Before performing apheresis, certain treatments must be discontinued to avoid interference with the procedure and to ensure optimal patient outcomes. The following guidelines should be adhered to:

- **Steroids:** Systemic steroid use should be discontinued at least 72 hours before apheresis. However, the physiological use of steroids, such as hydrocortisone ≤ 12 mg/m²/day (or equivalent), is permitted.
- **Chemotherapy and immunosuppressive drugs:**
  - The following medications should be stopped at least 1 week prior to apheresis and should not be administered concomitantly or after lymphodepleting chemotherapy:
    - Vincristine
    - 6-mercaptopurine
    - 6-thioguanine
    - Methotrexate (< 25 mg/m²)
    - Cytosine arabinoside (< 100 mg/m²/day)
    - Asparaginase (non-pegylated)
  - **Other systemic chemotherapy** must be discontinued at least 2 weeks before apheresis, except for the following drugs, which must be stopped at least 4 weeks prior:
    - Monoclonal antibodies (e.g., rituximab)
    - Pegylated asparaginase
    - Clofarabine
    - Fludarabine

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- **Hematopoietic Stem Cell Transplant (HSCT):** HSCT must be completed 12 weeks before apheresis.
- **GVHD Treatment:** Systemic treatments for graft-versus-host disease (GVHD) must be stopped at least 2 weeks prior to apheresis.
- **Blinatumomab:** Should be stopped at least 72 hours prior to apheresis.
- **Intrathecal Chemotherapy:**
    - Methotrexate: Should be discontinued at least 1 week before apheresis.
    - Cytarabine and/or steroids administered intrathecally are allowed up to 72 hours before infusion.
- **Tyrosine Kinase Inhibitors (TKIs):** Should be discontinued at least 72 hours before apheresis.

For any other treatments that have been recently administered or if there are uncertainties, please contact the coordinating investigator for further guidance.

7.5.2 Prohibited treatments prior to MB-CART2219.1 infusion

The use of steroids, chemotherapy, targeted therapies, and/or immunosuppressive drugs must follow the guidelines below, unless administered in a life-threatening emergency (e.g., steroids for severe CRS):

- **Steroids**: Systemic steroid use should be discontinued at least 72 hours before infusion. However, physiological doses of steroids (e.g., hydrocortisone ≤ 12 mg/m²/day or equivalent) are permitted.
- **Chemotherapy and immunosuppressive drugs**:
    - The following drugs must be discontinued at least 1 week prior to infusion and should not be used concomitantly with or following lymphodepleting chemotherapy:
        - Vincristine
        - 6-mercaptopurine
        - 6-thioguanine
        - Methotrexate (< 25 mg/m²)
        - Cytosine arabinoside (< 100 mg/m²/day)
        - Asparaginase (non-pegylated)
    - **Bridge therapy**: Systemic chemotherapy used as a bridging therapy prior to CAR-T infusion should be stopped at least **2 weeks** before infusion, except for monoclonal antibodies (e.g., rituximab) or pegylated asparaginase, which must be discontinued at least 4 weeks before infusion.
- **Intrathecal Chemotherapy**: Must be discontinued at least 1 week before CAR-T infusion.
- **Tyrosine Kinase Inhibitors (TKIs)**: TKIs must be stopped at least 72 hours before infusion.
- **Irradiation**:
    - Non-CNS Irradiation: Must be completed at least **2 weeks** prior to infusion.
    - CNS Irradiation: Must be completed at least **4 weeks** before infusion.

For any other treatments administered between apheresis and CAR-T infusion, or if further clarification is required, please contact the sponsor or coordinating investigator.

8 EFFICACY ASSESSMENT

8.1 Description of Efficacy endpoints assesment parameters

Complete remission (CR):

- Complete remission is assessed after haematopoietic recovery, usually at D28
- Complete remission requires to fulfil all the 4 following criteria:
    - bone marrow blasts <5%,

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- ANC >1 G/L,
- platelets > 100 G/L,
- and absence of extramedullary disease (CNS, other)

- Complete remission with incomplete haematological recovery (CRi) requires the same criteria than CR but with ANC ≤ 1 G/L or platelets ≤ 100 G/L

Complete remission with incomplete haematological recovery (CRi):

- Complete remission with incomplete haematological recovery (CRi) requires the same criteria as for Complete Remission (CR), with only one exception:
  - ANC: ≤ 1 G/L or
  - Platelets: ≤ 100 G/L

8.2 Definition of outcome variables

Overall Survival (OS) is defined as the time from enrolment to the death of any cause. Patients alive at their last follow-up will be censored.

Disease-Free Survival (DFS) is defined as the time from CR/CRi to disease progression or death. Patients alive in continuous CR at their last follow-up will be censored.

Duration of remission (DoR) is defined for patients in CR as the time from CR assessment until disease progression. Patients in continuous CR will be censored at their last follow-up.

8.3 Anticipated methods and timetable for measuring, and collecting the efficacy data

8.3.1 Methods for Measuring Efficacy Data

Efficacy will be assessed using a combination of clinical, laboratory, and imaging methods, depending on the primary and secondary endpoints of the study. The following methods will be employed to measure efficacy:

- **Complete Remission (CR) and Complete Remission with Incomplete Haematological Recovery (CRi)**:
  - Locally evaluated by bone marrow biopsy and aspiration, performed at predefined time points (e.g., Day 28, Month 3, Month 6, etc.)
  - MRD status will be locally assessed using quantitative PCR (qPCR) to monitor minimal residual disease.
- **Overall Survival (OS)**:
  - Time from enrolment to death from any cause will be recorded. Participants who are still alive at their last follow-up will be censored.
- **Augmented DFS (aDFS):**
  - Defined for patients in CR/CRi as the time from CR/CRi assessment until disease progression, introduction of a new treatment line based on MRD assessment, or CAR-T loss of persistence (loss of B-cell aplasia), or death from any cause. Patients alive in continuous CR will be censored at their last follow-up.
- **Loss of B-cell aplasia**
  - Return of normal circulating $CD19^+$ B-cells >50/µL sustained on two consecutive measures ≥14 days apart.

8.3.2 Timetable for Data Collection

Efficacy data will be collected at regular intervals throughout the study. The following timetable outlines key time points for data collection:

- Post-infusion Assessments**:**
  - Day 28**:** Primary efficacy assessment for CR and CRi.

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- Month 3, 6, 12, 18, and 24**:** Follow-up assessments to monitor disease progression, MRD status, and survival outcomes.

- Long-term Follow-up**:** Post 24-month assessments every 3 to 6 months to evaluate long-term survival, B-cell recovery, and ongoing CAR-T persistence.

8.3.3 Methods for Data Collection

- Electronic data capture (eCRF)**:** all clinical, laboratory, and imaging data will be recorded in an electronic Case Report Form (eCRF) for consistency and accuracy. Sites will be trained to ensure data collection follows the protocol.

- Imaging and laboratory data: Data from bone marrow biopsies, MRD assessments, and imaging will be uploaded and cross-checked for completeness and accuracy.

9 SPECIFIC STUDY COMMITTEES

9.1 Steering Committee

The steering committee (SC) is composed of coordinating investigators, the biostatistician, representatives of Clinical Research Unit (CRU), and sponsor representatives including the project manager of the study.

The aim of the SC is to coordinate information, to oversee studies, in particular with respect to algorithm guided doses escalation.

The DRCI sponsor retains decision-making authority.

10 SAFETY ASSESSMENT - RISKS AND BURDEN ADDED BY THE STUDY

10.1 Description of Safety endpoints assessment parameters

The primary endpoint for Phase 1 is the evaluation of DLT of MB-CART2219.1 within 28 days after infusion. The DLT is defined as any of the following adverse events (AEs) related to the IMP:

- **Any non-haematological, grade 4 or higher, clinically relevant adverse event** (AE), with the following exceptions:
  - Fever or infection
  - Laboratory abnormalities judged not to be clinically relevant
  - Events that can be promptly controlled by appropriate medical management
  - grade ≥ 4 TLS resolved within 7 days
- **Immune effector cell-associated neurotoxicity syndrome (ICANS)**
  - Any grade ≥ 4
  - Grade 3 that fails to resolve to grade ≤ 2 within 72 hours despite appropriate therapy
- **Tumor lysis syndrome (TLS)**
  - Any grade ≥ 4 that fails to resolve within 7 days
- **Cytokine release syndrome (CRS)**
  - Any grade ≥ 4
  - Grade 3 that fails to resolve to grade ≤ 2 within 72 hours despite appropriate therapy

All AEs will be graded according to CTCAE v6.0, with the exception of CRS and ICANS, which will be graded according to the American Society for Transplantation and Cellular Therapy (ASTCT) scale (see chapter 7.3).

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10.2 Anticipated methods and timetable for measuring, collecting and analysing the safety endpoints

Safety data will be measured through clinical, laboratory, and patient-reported assessments at predefined time points during the study. The following methods will be used:

- **Adverse Events (AEs)**:
  - Continuous monitoring of AEs from the time of informed consent until the end of the study.
  - AEs will be recorded and graded according to CTCAE v6.0 or, for CRS and ICANS, according to the ASTCT grading scale.
- **Laboratory Assessments**:
  - Routine haematology, biochemistry, and immunological testing will be performed at regular intervals to monitor for toxicity.
  - Specific assessments will include complete blood counts (CBCs), liver function tests (LFTs), renal function markers (creatinine, GFR), and cytokine levels (for CRS).
- **Vital Signs and Physical Examinations**:
  - Continuous monitoring of vital signs, including blood pressure, heart rate, temperature, and oxygen saturation will be conducted.
  - Physical exams will be performed at scheduled visits to assess general health and detect any signs of adverse reactions (e.g., neurotoxicity, infection).
- **Neurological Assessments**:
  - Frequent neurologic evaluations, including the Immune Effector Cell-Associated Encephalopathy (ICE) score, will be performed to monitor for signs of ICANS and other neurological toxicities.
- **Infection Surveillance**:
  - Routine screening for infections will include clinical assessment, **PCR testing** for viral reactivations (e.g., EBV, CMV), and cultures in case of suspected bacterial or fungal infections.
  - Prophylactic and therapeutic interventions will be recorded.
- **ICANS and CRS Monitoring**:
  - CRS and ICANS will be closely monitored, with daily assessments during the high-risk period (first 28 days post-infusion). Severity will be tracked using ASTCT criteria.
  - Laboratory markers (e.g., ferritin, CRP, IL-6 levels) will be measured to detect early signs of CRS.

10.2.1 Timetable for data collection (see chapter 5 for more details)

Safety data will be collected according to the following schedule:

- **Baseline**: Collected prior to apheresis, including complete medical history, baseline laboratory assessments, and performance status (ECOG score).
- **Pre-infusion**: Assessments will include vital signs, laboratory tests, and any treatment-related adverse events occurring after apheresis and prior to infusion.
- **Post-infusion (Day 0 - Day 28)**:
  - Daily monitoring of vital signs, clinical status, and laboratory values during hospitalization, particularly during the high-risk period for CRS and ICANS.
  - Neurologic assessments will be performed daily during this period.

After the DLT evaluation period, patients will be monitored according to following schedule:

- **Short-term Follow-up** (Month 1 - Month 3):

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    - Monthly safety assessments during the first month, including complete blood counts, infection surveillance, and organ function monitoring.
    - Monthly assessments for the next two months.
- **Long-term Follow-up** (Months 3 - 24):
    - Every 3 months, safety assessments will continue to monitor for long-term toxicities (e.g., delayed cytopenias, infections).
    - Patients will undergo routine laboratory tests and physical exams to detect any late-onset adverse events.
- Extended monitoring for long-term safety will continue for up to 15 years, in line with regulatory guidelines for gene therapy trials.

10.2.2 Methods for Data Collection

- Electronic Data Capture (eCRF): All safety data will be recorded in the electronic Case Report Form (eCRF). Sites will ensure timely and accurate entry of all data, including vital signs, laboratory values, and AE/SAE reports.
- Adverse Event Reporting: All adverse events, including SAEs and DLTs, will be documented and reported in real-time, as per regulatory requirements (see 10.3).
- Laboratory and imaging data: Laboratory results and any imaging data (e.g., brain MRI or CT for ICANS) will be uploaded to the eCRF and cross-checked for accuracy.

10.2.3 Methods for Data Analysis

- Descriptive Analysis: Safety data will be summarized using descriptive statistics, including the incidence, severity, and duration of AEs, SAEs, CRS, and ICANS.
- Data Safety Monitoring Board (DSMB): The DSMB will regularly review all safety data, particularly Grade ≥4 CRS and Grade ≥3 ICANS, and will provide recommendations regarding trial continuation or modifications.

10.2.4 Timetable for Data Analysis

Safety analysis will be conducted:

- after the first patient included in each dose level has completed the DLT evaluation (28 days post infusion),
- after the completion of each dose level, to assess the tolerability and any emerging safety signals according to the CRM algorithm

A final comprehensive safety analysis will be performed once all patients have completed the 28-day post-infusion safety follow-up and at the conclusion of the study.

10.3 Recording and reporting adverse events

10.3.1 Definitions

- **Adverse event AE**

Any untoward medical occurrence in a subject, to whom a medicinal product is administered and which does not necessarily have a causal relationship with this treatment.

- **Serious adverse event (SAE)**

Any untoward medical occurrence that at any dose requires inpatient hospitalisation or prolongation of existing hospitalisation, results in persistent or significant disability or incapacity, results in a congenital anomaly or birth defect, is life-threatening, or results in death.
Certain medical events may jeopardise the subject or may require an intervention to prevent a SAE, known as "important medical events" should also be considered as serious adverse events.

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- **Adverse reaction (AR)**

A response to a medicinal product which is noxious and unintended and which occurs at doses normally used in man for the prophylaxis, diagnosis or therapy of disease or for the restoration, correction or modification of physiological function. This definition also covers medication errors and uses not provided for in the protocol, including misuse or abuse of the product. In determining whether an adverse event is an adverse reaction, consideration shall be given to whether there is a reasonable possibility of establishing a causal relationship between the event and the investigational medicinal product based on an analysis of available evidence.

- **Suspected Unexpected serious adverse reaction (SUSAR)**

A serious adverse reaction, the nature, severity or outcome of which is not consistent with the reference safety information.

- **Important Medical Event**

Adverse event that may not be immediately life-threatening or result in death or hospitalisation but may jeopardise the subject or require medical or surgical intervention to prevent one of the outcomes listed under the definition of Serious Adverse Event. Examples of such events include invasive or malignant cancers, intensive treatment (in an emergency room or at home) for allergic bronchospasm, blood dyscrasias or convulsions that do not result in hospitalisation, or the development of drug dependency or abuse, overdose or misuse.

- **Unexpected event**

An unexpected event which affects the benefit-risk balance of the clinical trial, but are not suspected unexpected serious adverse reactions as referred to in Article 42 of Regulation (EU) No 536/2014 of the European parliament. That notification shall be made without undue delay but no later than 15 days from the date the sponsor became aware of this event.

- **Urgent safety measure**

Where an unexpected event is likely to seriously affect the benefit-risk balance, the sponsor and the investigator shall take appropriate urgent safety measures to protect the subjects.
The sponsor shall notify the Member States concerned, through the EU portal, of the event and the measures taken.
That notification shall be made without undue delay but no later than seven days from the date the measures have been taken.

- **Serious breach**

A 'serious breach' means a breach likely to affect to a significant degree the safety and rights of a subject or the reliability and robustness of the data generated in the clinical trial.

NB: The sponsor is responsible for determining whether the situation corresponds to a serious breach. The sponsor must record and document the serious breach, analyse its fundamental causes, record the information in the CTIS within the time scale described in the Clinical Trial Regulation and take corrective and preventive measures where relevant. In the event of a serious breach likely to affect the benefit/risk ratio of the clinical trial, the notification in the CTIS may be accompanied by other notifications such as those relating to an unexpected event with an impact on the benefit/risk ratio of the clinical trial, urgent safety measures or a declaration of an unexpected serious adverse reaction (SUSAR) for example.

- **Medication error**

The unintentional omission or commission of an action during a treatment process involving a medicinal product that may cause a risk (potential error) or an adverse reaction in the patient.

A medication error may be:

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a. Proven and cause an adverse reaction (error with adverse drug reaction) or not cause an adverse reaction (error without harm)
b. Intercepted before administration to the patient (intercepted error)
c. A potential error exists if an observation identifies a potential danger to the patient.

- **Abuse of a medicinal product**

Persistent or sporadic, intentional excessive use of one or more psychoactive substances which is accompanied by harmful physical or psychological effects.

- **Misuse**

Intentional use that does not comply with the clinical trial framework for an inappropriate medical purpose.

- **Overdose**

Administration of a quantity of a medicinal or other product, given per administration or cumulatively, which is above the maximum recommended dose according to the summary of product characteristics. In practice this is an exposure resulting in high plasma concentrations. It may involve the excessive taking (intentional or accidental) of a medicinal product.

10.3.2 The role of the investigator

The investigator must **assess the seriousness of each adverse event** and record all serious and non-serious adverse events and all special circumstances (medication errors, in utero exposure / pregnancy / breastfeeding, uses outside what is foreseen in the protocol, including misuse and abuse of the product) in the case report in the case report form (eCRF).

For all serious adverse events, the investigator **must also notify the SAE to the safety department** as soon as possible, except SAE not requiring a notification without delay described in section 10.3.3.1.2. "Serious adverse events that do not require the investigator to notify the sponsor without delay".

The investigator must **document** serious adverse events **as thoroughly as possible** and provide a definitive medical diagnosis, if possible.

The investigator must **assess the intensity** of the adverse events using the scale for grading adverse events appended to the protocol:

- ASTCT grading scale for specific CAR-T cells toxicity (CRS and ICANS)
- CTCAE v.6 for non-specific toxicities

The investigator must **assess the causal relationship** between serious adverse events and

- the IMP
- and/ or interventions/procedures added by the study (blood samples, brain MRI at baseline).
- and/or auxiliary medicinal products (cf. section 7.3)
- and/or interaction between auxiliary medicinal product and the IMP

Two situations are possible for auxiliary medicinal products:
1 - In case of adverse reactions only related to authorised auxiliary medicinal product(s) in accordance with Regulation (EC) No 726/2004 or in any Member State concerned in accordance with Directive 2001/83/EC: SAE must be notified to sponsor's safety department as soon as possible and also the investigator must report to the relevant regional pharmacovigilance center (Centre Régional de Pharmacovigilance (CRPV)) or to www.signalement-sante.gouv.fr.
2 - In case of a suspected interaction between authorised auxiliary medicinal product(s) and the investigational medicinal product(s), the reporting rules for the IMP apply: SAE must be notified to the sponsor's safety department as soon as possible.

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The method to be used by the investigator is based on 2 causality terms (EudraVigilance Clinical Trial Module (EVCTM) method):

- Related
- Not related

10.3.3 Serious adverse events

The investigator notifies the sponsor without undue delay but not later than within 24 hours on the day the investigator becomes aware of any serious adverse event which occurs during a study that meets the description in article 41 of Regulation (EU) N°536/2014, with the exception of any event which is listed in the protocol and, if applicable, in the investigator's brochure as not requiring notification.

A serious adverse event is any untoward medical occurrence that:

1- results in death
2- is life-threatening to the participant enrolled in the study
3- requires hospitalisation or prolongation of existing hospitalisation
4- results in persistent or significant disability/incapacity
5- is a congenital anomaly/birth defect
6- important medical event

Moreover, special circumstances require the investigator to notify the sponsor without delay are:
-medication errors,
- *in utero* exposure / pregnancy / breastfeeding
- uses outside what is foreseen in the protocol, including misuse and abuse of the product.

All pregnancies must be reported to the safety department by the investigators within 24 hours on the day the investigator becomes aware of it.

Only special circumstances associated to an SAE must be reported to the safety department by the investigators and must be recorded in the "SAE" module of the CRF.

Special circumstances not associated to an SAE must be reported to the CRF in the "AE" module by the investigators and notify as a deviation to the protocol by the clinical trial unit.

10.3.3.1 Specific features of the protocol

<u>Other events that require the investigator to notify without delay the sponsor</u>:

- **Adverse events particularly followed by the sponsor** for the safety and assessment:
    - CRS, IEC-HS
    - ICANS
    - TLS

  These events must be notified by the investigator to the sponsor without delay on the day the investigator becomes aware of these adverse events, in the same manner and within the same deadline as for serious adverse events (see above).
  For these adverse events particularly followed by the sponsor, please, check the "important medical event" box as a seriousness criterion in the SAE notification form.

- ***In utero* exposure**

  The investigator must notify the sponsor without delay on the day the investigator becomes aware of any pregnancy that occurs within 2 years after IMP infusion, even if it is not associated with an adverse event.
  If the investigational medicinal product is genotoxic, every case of maternal or paternal exposure must be notified.

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Notify using a specific form appended to the protocol.

- **Exposure while breastfeeding**

  Exposure while breastfeeding occurs if an infant or a child may have been exposed to a medicinal product *via* the breast milk of a mother being treated with an investigational medicinal product. **Even if it is not associated with an adverse event, the investigator must notify the sponsor without delay on the day the investigator becomes aware of the exposure while breastfeeding.**

10.3.3.2 Serious adverse events that do not require the investigator to notify the sponsor without delay:

These serious adverse events are only recorded in the case report forms.
A periodic extraction of all such SAEs from the e-CRF will be performed by the Clinical Trial Unit and emailed to the Safety Department at the same cadence as DSMB reviews (i.e., after each cohort of 3 patients) and at least once per year for the annual safety report. Extractions will be sent to: expertisecsi.drc@aphp.fr

- Hospitalisation for routine supportive care of underlying B-ALL (red cell/platelet transfusions for cytopenias; bridging therapy before to start lymphodepletion)
- Hospitalisation while the patient is in CR/CRi (e.g., for symptomatic anaemia, antimicrobial therapy for low-grade, uncomplicated infections managed per local standard)
- Hospitalisation or prolongation of hospitalisation for organisational/logistical/psychosocial reasons, without new clinical deterioration

The following circumstances do not require the investigator to notify the sponsor without delay:

- Hospitalisation for a pre-existing illness or condition
- Hospitalisation for a medical or surgical treatment scheduled prior to the study
- Admission for social or administrative reasons
- Medication errors or uses outside what is foreseen in the protocol, including misuse and abuse of the product without SAE and not in relationship with a serious breach

10.3.4 Period during which SAEs must be notified without delay by the investigator to the sponsor

The investigator must notify the sponsor immediately of any serious adverse event as defined in the corresponding section:

- from the date on which the participant signs the consent form (screening visit)
- throughout the whole follow-up period up to 2 years after IMP infusion, or 1 month after relapse, whatever occurs first.
- Indefinitely, if the SAE is likely to be due to the investigational medicinal product or to the study interventions (e.g., serious reactions that could appear long after exposure to the medication, such as cancers or congenital abnormalities) *

*NB: In this last case, the investigator **does not have to collect indefinitely in the case report form (CRF or eCRF)** all **SAEs possibly related** to the clinical trial, **but must notify them**, to the sponsor, **as soon as he/she becomes aware of them, by the SAE notification form or by email or by fax (as described below)**.

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10.3.4.1 Procedures and deadlines for notifying the sponsor

The initial notification of an SAE must be provided in a written report signed by the investigator using a SAE notification form specific to the study and intended for this purpose (in the case report form).

Each item in this form must be completed by the investigator so that the sponsor can carry out the appropriate analysis.

The initial notification to the sponsor of a serious adverse event must be quickly followed by an additional detailed written report (or reports) so that the case outcome may be monitored by the Safety Department or to provide further information.

Whenever possible, the investigator will provide the sponsor with any documents that may be useful (medical reports, laboratory test results, results from additional examinations, etc.). These documents must be non-identifying. In addition, the documents must include the following: study acronym, number and participant's reference number.

Any adverse event will be monitored until fully resolved (stabilisation at a level considered acceptable by the investigator, or return to the previous state) even if the participant has left the study.

The initial notification, the SAE follow-up reports and all other documents must be sent to the sponsor's Vigilance Department by email (eig-vigilance.drc@aphp.fr). It should be noted that it is possible to send SAE reports to the Safety Department by fax to +33 (0)1 44 84 17 99, only in the event of a failed attempt to send the SAE report by email (in order to avoid duplication).
When sending the email, please:

- Adopt a standardized nomming of the email subject in the following form:

<< Objet : YYYYYY_XXXXXX_jjmmaaaa (avec YYYYYY : code de la recherche, XXXXXX : acronyme de la recherche et jjmmaaaa : date de transmission). >>

- Send a SAE initial notification form and/or a follow-up report concerning a single participant for a given SAE, attachment may contain one (or more) document(s) (follow-up and hospitalization reports, for example).
The total size of the email must be less than 8 MB, otherwise please send several e-mails.

- Ensure that all documents transmitted (e.g. hospital reports) are anonymized and identified with the participant's identification number.

The investigator should follow the procedure:

- the investigator completes the SAE notification form in the e-CRF, then validates, prints, and signs the form before sending it by email;
- if it is not possible to connect to the e-CRF, the investigator should complete, sign, and send the SAE notification form to the Safety Department. As soon as the connection is restored, the SAE notification form in the e-CRF must be duly completed.

The investigator must respond to all requests from the sponsor for additional information (new information must be also recorded in the e-CRF).
For all questions relating to the notification of an adverse event, the Safety Department can be contacted via email: vigilance.drc@aphp.fr.

For cases of *in utero* exposure, the investigator will complete the Notification and Follow-up form for a pregnancy occurring during participation in a study".
The investigator must monitor the pregnant woman throughout her pregnancy or until the pregnancy is terminated, and must notify the sponsor of the outcome of the pregnancy using this form.

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If the outcome of the pregnancy falls within the definition of a serious adverse event (miscarriage, pregnancy termination, foetal death, congenital abnormality, etc.), the investigator must follow the procedure for reporting SAEs.
The initial pregnancy notification, the SAE follow-up reports, and any other documents will be sent to the sponsor according to the same procedures specified herein.

10.3.5 Role of the sponsor

The sponsor, represented by its Safety Department, shall continuously, throughout the trial assess the safety of the IMP throughout the study.

10.3.5.1 Analysis and declaration of serious adverse events

The sponsor assesses:

- the **seriousness** of all the adverse events reported
- the **causal relationship** between these events and the IMP, the auxiliary medicinal products, the interventions added by the study and any concomitant treatments,
  All serious adverse events which the investigator and/or the sponsor reasonably believe may have a causal relationship with the IMP are classed as suspected serious adverse reactions.
- the **expected or unexpected nature** of the serious adverse reactions related to IMP and to the interventions, procedures or examinations specific to the study.
  Any serious adverse reaction whose nature, severity, frequency or outcome is inconsistent with the safety information described in the summary of product characteristics, or in the investigator's brochure if the product is not authorised, is considered unexpected. The sponsor, represented by its Safety Department, assesses the expected/unexpected nature of a serious adverse reaction based on the information described below**.**

  For serious adverse events likely to be related to the IMP: refer to the section “Reference safety information” of the Investigator's Brochure uploaded to CTIS platform and enclosed in the investigator study file.

❖ For serious adverse events that maybe related to the auxiliary medicinal products, refer to the SmPC for “allopurinol”, “cyclophosphamide”, “fludarabine”, “febuxostat”, “rasburicase”, “tocilizumab”, “dexamethasone”, “anakinra”, “ruxolitinib”, “Ig IV” mentioned in the cover letter and enclosed in the investigator study file.

❖ The serious adverse events potentially related to the interventions, procedures or examinations specific to the study are:
- related to blood samples: hematoma, pain, vagal malaise, mild bleeding
- related to brain MRI w/contrast: claustrophobia (anxiety, feeling of being closed, panic, sweating and difficulty breathing), anxiety attack, nausea, vomiting, panic attack, dizziness, malaise without traumatic consequences, hypersensitivity reactions (including rash, urticaria, bronchospasm, or anaphylaxis), injection site complications (extravasation, local pain, inflammation
- related to leukapheresis: fatigue, dizziness, complications related to central venous access

The sponsor will report all suspected unexpected serious adverse reactions (SUSARs), via Eudravigilance within the regulatory time frame, to the competent authority:

- in the case of fatal or life-threatening suspected unexpected serious adverse reactions, as soon as possible and in any event not later than 7 days after the sponsor became aware of the reaction;
- in the case of non-fatal or non-life-threatening suspected unexpected serious adverse reactions, not later than 15 days after the sponsor became aware of the reaction;

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- in the case of a suspected unexpected serious adverse reaction which was initially considered to be non-fatal or nonlife threatening but which turns out to be fatal or life-threatening, as soon as possible and in any event not later than 7 days after the sponsor became aware of the reaction being fatal or life-threatening
- in the uncompleted case of fatal or life-threatening suspected unexpected serious adverse reactions, all additional information to complete this case have to be reported as soon as possible and, in any event, not later than 8 days after the initial report
- All additional, relevant information must be declared by the sponsor in the form of follow-up reports within a period of 15 days starting from when the sponsor had this information.

The sponsor must notify all the investigators involved about any information that could adversely affect the safety of the research participants.

10.3.5.2 Analysis and declaration of other safety data

This means any new information that prompts a reassessment of the risk/benefit ratio of the trial or of the product under investigation, or which could be sufficient to consider changes to the use of the product, the trial procedures or trial documentation, or the suspension, cancellation or amendment of the research trial or of other similar trials.

The sponsor will report in CTIS platform without delay upon knowledge of any emerging safety issue and, if applicable, describe what measures have been taken.

The unexpected event shall be notified without undue delay but no later than 15 days from the date the sponsor became aware of this event.

The urgent safety measures shall be notified without undue delay but no later than 7 days from the measures have been taken.

Following the initial declaration of any emerging safety issue, the sponsor will address to competent authorities any additional relevant information about the emerging safety issue in a follow-up report.

10.3.5.3 Annual safety report

Once a year for the duration of the clinical study, the sponsor must produce an annual safety report (Development Safety Update Report - DSUR) which includes, in particular:

- a safety analysis for the research participants,
- a description of the patients included in the study (demographic profile etc.)
- a list of all the suspected serious adverse events that occurred during the period covered by the report,
- summary tables of all the serious adverse events that have occurred since the start of the study,

The sponsor produces one annual safety report (Development Safety Update Report - DSUR) for one clinical trial.

The report must be submitted in CTIS no later than 60 days after the anniversary of the date on which the competent authority authorised the trial.

The final annual safety report must be submitted in CTIS no later than 60 days after the end date of the clinical trial.
The end date of the clinical trial is defined as the date of the last visit of the last subject.

10.3.6 Data Safety Monitoring Board (DSMB)

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The Data Safety Monitoring Board (DSMB) may be established by the sponsor. Its primary mission is to monitor safety data. It can have other missions, such as monitoring efficacy data (especially if the protocol includes interim analyses).
The sponsor is responsible for justifying the creation or absence of such a committee to the Competent Authority (ANSM) and to the CCP (Research Ethics Committee).

A DSMB will be established for this clinical trial. The DSMB will hold its preliminary meeting before the first inclusion of the first participant and ideally before the protocol is submitted to the competent authority and the Ethics Committee.

All missions as well as the specific operating procedures of the DMC are described in the study's DSMB charter.

The members of the DSMB are:
- Marie-Emilie Dourthe
- Tony Marchand
- Xavier Paoletti

The DSMB will operate in accordance with the sponsor's procedures. The DSMB works in an advisory capacity only and the sponsor retains all decision-making authority.

11 DATA MANAGEMENT

11.1 Data collection procedures

The investigator will allow the sponsor's representatives to monitor the study at the frequency defined in the contract, depending on enrolment at each center.
Case Report Forms (CRFs) and related source documents will be reviewed in detail during monitoring visits (completeness, adherence to the guidelines, accuracy compared to source documents). The sponsor's representative will also review regulatory documents, drug storage and accountability. The investigator must keep a comprehensive and centralized filing system for all study-related documents that would be suitable for inspection by the sponsor's monitors or representatives of any other regulatory agency.

11.2 Identification of data recorded directly in the CRFs which will be considered as source data

Not applicable

11.3 Right to access data and source documents

11.3.1.1 Data access

In accordance with GCPs and appendix 1 of the European Regulation N°536-2014:
- the sponsor is responsible for ensuring all parties involved in the study agree to guarantee direct access to all locations where the study will be carried out, the source data, the source documents and the reports, for the purposes of the sponsor's quality control and audit procedures or inspections by the competent authority.
- the Sponsor declares that investigators and participating institution will ensure the persons in charge of monitoring, quality control and auditing or inspecting the clinical trial have access to the documents

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and personal data strictly necessary for these tasks, in accordance with the statutory and regulatory provisions in force

11.3.1.2 Source documents

Source documents are defined as any original document or item that can prove the existence or accuracy of a data or a fact recorded during the study. These documents will be kept in accordance with the regulations in force by the investigator or by the hospital in the case of a hospital medical file.

11.3.1.3 Data confidentiality

The persons responsible for the quality control of clinical studies will take all necessary precautions to ensure the confidentiality of information relating to the IMP, the study, the study participants and in particular their identity and the results obtained.
These persons, as well as the investigators themselves, are bound by professional secrecy.
During and after the clinical trial, all data collected concerning the participants and sent to the sponsor by the investigators (or any other specialised collaborators) will be rendered non identifying.
Under no circumstances shall the names and addresses of the participants involved be shown.
Only the participant's initials will be recorded, accompanied by an encoded number specific to the study indicating the order of enrolment.
The sponsor will ensure that each participant has given written permission for any personal information about him or her which is strictly necessary for the quality control of the study to be accessed.

11.4 Data processing and storage of research documents and data

3. Identification of the data processing manager and location(s)

Data processing will be performed by Pr Jérôme Lambert in the « Service de Biostatistiques et Information Médicale (SBIM), Hôpital Saint-Louis, Paris » and in URC Lariboisère Saint-Louis, Saint-louis Hospital Data entry.

4. Data entry

Non-identifying data will be entered electronically via a web browser.

11.5 Data ownership

AP-HP is the owner of the data. The data cannot be used or disclosed to a third party without its prior permission.

12 STATISTICAL ASPECTS

12.1 Description of statistical methods to be used including the timetable for the planned interim analyses

12.1.1 Safety and Efficacy Analyses

Safety and primary efficacy analyses will be conducted on all patients treated with MB-CART2219.1.

Regarding secondary endpoint such as OS a secondary analysis will be performed on the whole population of patients having received apheresis regardless of infusion.

Phase 1 Safety Analysis:

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Data in Phase 1 will be recorded sequentially according to the Continual Reassessment Method (CRM). A parametric model will be employed, assuming a monotonic relationship between dose level and DLT. The model will estimate the DLT rate at each dose level.
It requires statistical parameters to be defined

- Choice of the dose levels (K=4)
    - **Dose Level (DL) 0**: 0.25 x 10^6 CAR-T cells/kg, starting dose
    - **Dose Level (DL) 1**: 0.5 x 10^6 CAR-T cells/kg
    - **Dose Level (DL) 2**: 1.0 x 10^6 CAR-T cells/kg
    - **Dose Level (DL) 3**: 2.0 x 10^6 CAR-T cells/kg

- Definition of the DLT target rate, θ =15%.
- Choice of the working model; as most commonly used, the model choice implementation in will use the "empiric" model to model the DLT probabilities R(d k) at each dose
  $R(d_k) = Pr(DLT\ at\ dose\ d_k) = \alpha_k^{exp(a)}$
  Where $(\alpha_k)$is the pre-specified skeleton of the working model and a is the model parameter to be estimated by the data.
- Skeleton choice: it will rely on the algorithm of Lee and Cheung (Lee SM, Cheung YK. Model calibration in the continual reassessment method. Clinical Trials. 2009;6:227–38.). It is available as a function, getprior, within the R package dfcrm and requires four pieces of information in order to generate the skeleton; the prior MTD (v=DL2), the target toxicity rate (θ=15%), the number of dose levels (K=4), and a spacing measure (δ=0.05, as recommended) of the skeleton.
- Noninformative prior: As recommended for the empiric model a mean zero normal prior $N(0, \sigma_a^2)$ will be used. The prior variance, σ_a^2, that completely specifies the prior distribution, will be set as the least informative normal prior using the calibration method described in (Lee SM, Cheung YK. Calibration of prior variance in the Bayesian continual reassessment method. Stat Med. 2011;30:2081–9.)
- The parameter estimation will be based on recorded data, using Bayesian inference to allow borrowing of information across dose levels throughout the trial, along the inclusion of cohort of 3 patients.
- Pre-trial simulation: Before inclusion, model characteristics will be evaluated using a pre-trial simulation study to validate assumptions. All parameters will be set and reported in the detailed Statistical Analysis Plan (SAP) before trial onset.
- Other design specifications: Two other important design specifications are to be defined
  1. Dose escalation skipping restriction: The trial is not allowed to skip dose levels when escalating.
  2. Stopping rule for safety: The trial will stop for safety if the lowest dose is indicated by the data to be too toxic. Specifically, with the procedure used in the application, the trial stops for safety if the lower limit of a 90% probability interval exceeds the target DLT rate.
- Software: The dfcrm R package (Cheung YK. Dose Finding by the Continual Reassessment Method. Chapman & Hall/CRC Biostatistics Series: Taylor and Francis; 2011) will be used for the analysis.

Phase 2 Efficacy Analysis:

- Intent-to-Treat (ITT)**:** The primary efficacy analysis will be conducted on an intent-to-treat basis. The analysis will provide a 95% confidence interval for all efficacy endpoints.

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- All missing data on the outcome will be considered as a failure.
- Secondary outcomes
  All survival data will be analysed with methods for right-censored time-to-failure data, including Kaplan-Meier estimate or nonparametric cumulative incidence function estimate in case of informative censoring.

12.1.2 Additional sensitivity analysis using external data

Since an academic dose finding trial is ongoing using the same MB-CART2219.1 but in a different population, a sensitivity analysis will be performed on the pooled results to ensure robustness of the safety findings.

12.1.3 Timetable for Interim Analyses

Phase 1 Interim Analysis:

- Safety analysis will be conducted **sequentially**, with an interim safety assessment after the **first patient** in each dose level has completed the DLT evaluation period (28 days post-infusion).
- An additional interim analysis will be performed **after each dose level** is completed to assess cumulative safety data and guide dose escalation decisions.

Phase 2 Interim Analysis:

- No interim analysis is planned for phase 2.

Final Analysis:

- A final analysis will be performed after the completion of the trial covering all enrolled patients and assessing safety, efficacy across all cohorts.

12.2 Calculation hypotheses for the number of participants required and the result

A one-stage Bayesian continual reassessment method (CRM) will be used with cohort size of 3. A maximum sample size of 18 ensures a probability of correct select of 0.5 for the CRM, assuming a 1.8 odds ratio for the true logistic dose-toxicity curve and using a basic arbitrary skeleton with DL2 as the anticipated MTD1. The skeleton and prior variance of the dose-toxicity model parameter will be further calibrated by simulation2-3, to optimize the probability of correct selection (PCS), accounting for the trial design and additional safety rules (see section 12.1), and detailed in the SAP before trial initiation. The sample size will likely allow convergence of dose assignments on the estimated MTD and result in an expanded sub-sample of patients assigned to the estimated MTD.

1: Cheung YK. Sample size formulae for the Bayesian continual reassessment method. Clin Trials Lond Engl. 2013;10(6):852-861. doi:10.1177/1740774513497294
2: Lee SM, Cheung YK. Calibration of prior variance in the Bayesian continual reassessment method. Stat Med. 2011;30(17):2081-2089. doi:10.1002/sim.4139
3: Lee SM, Cheung YK. Calibration of prior variance in the Bayesian continual reassessment method. Stat Med. 2011;30(17):2081-2089. doi:10.1002/sim.4139

12.3 Anticipated level of statistical significance

All statistical tests will be 2-sided, with p-values of 0.05 or less denoting statistical significance.

12.4 Statistical criteria for termination of the study.

Phase 1 could be discontinued due to an over-toxicity profile, based on the safety stopping rule described above (section 12.2).

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12.5 Method for taking into account missing, unused or invalid data

Missing data on the phase 2 primary outcome will be imputed as a failure.

12.6 Management of modifications made to the analysis plan for the initial strategy.

All modifications will be notified in an amendment of the statistical analysis plan.

12.7 Choice of individuals to be included in the analyses

All analyses will use the intent-to-treat population, regardless of protocol adherence.

Regarding phase 2, the main analysis will be performed on all patients treated at the selected dose level

A sensitivity analysis will be performed on all patients included in phase 1 + phase 2 regardless of the assigned dose level

13 QUALITY CONTROL AND ASSURANCE

Every research project involving human participants managed by AP-HP is ranked according to the projected risk incurred by study participants using the classification system specific to AP-HP sponsored interventional research involving human participants.

13.1 General organisation

The sponsor must ensure the safety and respect of individuals who have agreed to participate in the study. The sponsor must implement a quality assurance system to best monitor the implementation of the study in the investigation centres.

For this purpose, the sponsor shall assign Clinical Research Associates (CRA) whose primary role is to carry out regular follow-up visits at the study locations, after having carried out the initial visits.
The purpose of monitoring the study, as defined in the Good Clinical Practices (GCP section 5.18.1), is to verify that:

- the rights, safety and protection of the research participants are met
- the data reported are exact, complete and consistent with the source documents
- the study is carried out in accordance with the protocol in force, the GCPs and the statutory and regulatory provisions in force

5. Strategy for centre opening

The strategy for opening the centres established for this study is determined using the appropriate monitoring plan. The opening of the different participating centers will be performed on site or remotely (after risk assessment and analysis) by the CRA from the URC-DRCI from Saint-Louis hospital.

6. Scope of centre monitoring

In the case of this D risk study**,** the appropriate monitoring level has been determined based on the complexity, the impact and the budget for the study. Therefore, in agreement with the coordinating investigator, the sponsor has determined the logistical score and impact, resulting in a study monitoring level to be implemented: High risk level.
These various levels are defined in the monitoring charter for research involving human participants

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13.2 Quality control

A Clinical Research Associate (CRA) appointed by the sponsor will be responsible for the good completion of the study, for collecting, documenting, recording and reporting all handwritten data, in accordance with the Standard Operating Procedures applied within the DRCI (Clinical Research and Innovation Department) and in accordance with Good Clinical Practices as well as the statutory and regulatory requirements.

The investigator and the members of the investigator's team agree to make themselves available during regular Quality Control visits carried out by the Clinical Research Associate. During these visits, the following elements will be reviewed depending on the monitoring level:

- written consent
- compliance with the study protocol and with the procedures defined therein
- quality of the data collected in the case report form: accuracy, missing data, consistency of the data with the "source" documents (medical files, appointment books, original copies of laboratory results, etc.)
- management of the treatments used

13.3 Case report forms

Electronic CRF
All information required by the protocol must be entered in the case report forms. The data must be collected as and when they are obtained, and clearly recorded in these case report forms. Any missing data must be coded.
Every site will have access to the electronic case report forms via a web-based data collection system. Investigators will be given a document offering guidance on using this tool.
When the investigators complete the case report form via the Internet, the CRA can view the data quickly and remotely. The investigator is responsible for the accuracy, quality and relevance of all the data entered. In addition, the data are immediately verified as they are entered, thanks to consistency checks. To this end, the investigator must validate any changes to the values in the case report form. An audit trail will be kept of all changes. A justification can be added when applicable, as a comment.
A print-out, authenticated (signed and dated) by the investigator, will be requested at the end of the study. The investigator must archive a copy of the authenticated document that was issued to the sponsor.

13.4 Management of non-compliances

Any events that occur as a result of non-compliance – by the investigator or any other individual involved in running the study – with the protocol, standard operating procedures, good clinical practices or statutory and regulatory requirements must be recorded in a declaration of non-compliance and sent to the sponsor.
These non-compliances will be managed in accordance with the sponsor's procedures.
The sponsor shall notify the Member States concerned about a serious breach of this Regulation or of the version of the protocol applicable at the time of the breach through the EU portal without undue delay but not later than seven days of becoming aware of that breach.

13.5 Audits/inspections

The investigators agree to consent to the quality assurance audits carried out by the sponsor as well as the inspections carried out by the competent authorities. All data, documents and reports may be subject to regulatory audits and inspections. These audits and inspections cannot be refused on the grounds of medical secrecy.

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An audit can be carried out at any time by individuals appointed by the sponsor and independent of those responsible of the research. The aim of the audits is to ensure the quality of the study, the validity of the results and compliance with the legislation and regulations in force.

The persons who manage and monitor the study agree to comply with the sponsor's requirements and with the competent authority regarding study audits or inspections.

The audit may encompass all stages of the study, from the development of the protocol to the publication of the results, including the storage of the data used or produced as part of the study.

13.6 Principal Investigator's commitment to assume responsibility

Before starting the study, each investigator will give the sponsor's representative a copy of his/her updated personal *curriculum vitæ*, signed and dated less than one year, with his/her RPPS number (*Répertoire Partagé des Professionnels de Santé*, Collective Database of Health Professionals). The CV must include any previous involvement in clinical research and related training.
Each investigator will commit to comply with legislation and to conduct the study in line with GCP, in accordance with the Declaration of Helsinki.
The Principal Investigator at each participating site will sign a commitment of responsibility (standard DRCI document) which will be sent to the sponsor's representative.
The investigators and their staff will sign a delegation of duties form specifying each person's role and will provide their CVs.

13.7 Suitability of the facilities

Before starting the study, each clinical trial sites will give the sponsor's representative a duly justified written statement on the suitability adapted to the nature and use of the IMP and including a description of the suitability of facilities, equipment, human resources and description of expertise.

13.8 Pharmacist's commitment of responsibility

The pharmacist designated at each participating site undertakes to ensure the proper management of the IMP MB-CART2219.1, in accordance with Good Manufacturing Practice (GMP), the current regulatory framework for Advanced Therapy Medicinal Products (ATMPs), and the procedures defined in this protocol.

The pharmacist is specifically responsible for:

- Verifying the receipt, storage, traceability, and accountability of the IMP (fresh or cryopreserved product), including compliance with transport and conservation requirements.
- Ensuring that all quality controls, release documents, and chain of custody documentation accompanying the IMP are complete and verified before delivery for infusion.
- Coordinating with the centralized manufacturing platform (MEARY, Hôpital Saint-Louis) and the clinical team to guarantee synchronization between lymphodepletion and infusion, in line with the "Coordination Form" process described in Section 5.1.
- Preparing the IMP for administration, including thawing (if applicable), manipulation under controlled conditions, and release to the clinical team for infusion, in accordance with institutional SOPs.
- Maintaining comprehensive records of IMP receipt, preparation, dispensing, and destruction, and transmitting this documentation to the sponsor upon request.
- Reporting without delay any deviation, non-compliance, or product defect that could impact patient safety or trial integrity.

14 ETHICAL AND LEGAL CONSIDERATIONS

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14.1 Methods for informing research participant and obtaining their consent

According to article 29 of European regulation N°536/2014, No clinical trials on medicinal products for human use can be carried out on a person without his/her freely given and informed consent, obtained in writing after the person has been given the information specified in article 29.2 of the European Regulation
A reflection period of at least one day is given to the individual between the time when he or she is informed and when he or she signs the consent form.
The person's freely-given, written, informed consent will be obtained by the principal investigator or a physician representing the investigator, before the person is enrolled in the study.
A copy of the patient information sheet and consent form, signed and dated by the research participant and by the principal investigator or the physician representing the investigator will be given to the individual prior to their participation in the study. The principal investigator or the physician representing him/her will keep a copy.
At the end of the study, one copy will be placed in a tamper-proof sealed envelope containing all the consent forms. This envelope will be archived by the sponsor.
In addition, the investigator will specify in the person's medical file the person's participation in the research, the procedures for obtaining his/her consent or consent from any other person in the cases set forth by article European regulation N°536/2014 (art. 29 and following) as well as the methods used for providing information for the purpose of collecting it. The investigator will retain one copy of the signed and dated consent form.

14.2 Prohibition from participating in another clinical study or exclusion period set after the study

An exclusion period of participation after the participant has finished this study is defined in the context of this research. It will last for 15 days. This duration is justified by the severity of relapsed/refractory acute lymphoblastic leukemia.

The participant may not enrol in another interventional study for the duration of his or her participation without consulting with the physician monitoring him or her in the context of the study.

The participants can however participate in other non-interventional studies

14.3 Compensation for participants

There is no compensation in this study.

14.4 Authorisation for the research location

In France, the study will be carried out in treatment units on individuals presenting a clinical condition for which the units are specialised and who require interventions that are routinely performed at those units. Therefore, the research location does not require specific authorisation.

14.5 Legal obligations

7. 14.5.1. Role of the sponsor

Assistance Publique - Hôpitaux de Paris (AP-HP) is the sponsor of this study and, by delegation, the DRCI (Clinical Research and Innovation Department) carries out the study's missions in accordance with regulation (EU) No 536/2014 of the European Parliament and of the council of 16 April 2014 and all national Laws. Assistance Publique - Hôpitaux de Paris reserves the right to halt the study at any time for medical or administrative reasons. In this case, notification will be sent to the investigator.

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8. Request for authorisation

Prior to starting the study, AP-HP, as sponsor, must obtain authorisation from the Competent Authority and the Research Committee for this clinical trial on medicinal product for human use within the scope of its authority and in accordance with in force legislation and regulatory requirements.

9. Procedures relating to data protection regulations

The computer file used for this research is implemented in accordance with French (amended "Informatique et Libertés" law governing data protection) and European (General Data Protection Regulation – GDPR) regulations.
This Clinical trial is governed by the CNIL (French Data Protection Agency) "Reference Methodology for processing personal data used within the scope of health research" (amended MR-001). AP-HP, as sponsor of the research, has signed a declaration of compliance with this "Reference Methodology"

10. Start of the Clinical Trial

The sponsor shall notify each Member State concerned of the start of a clinical trial and the start of the recruitment through the EU portal. That notification shall be made within 15 days from the start of the clinical trial.

11. Amendments to the clinical trial

Any substantial modification to the protocol by the coordinating investigator must be sent to the sponsor for approval. After approval is given, the sponsor must obtain, approval from the Ethics Committee and authorisation from the competent authority within the scope of their respective authorities, before the amendment can be implemented.
The information note and the consent form can be revised, if necessary, in particular if there is a substantial amendment to the study or if adverse reactions occur.

12. End of the Clinical Trial

The end of the Clinical Trial corresponds to the end of the participation of the last person who participate to the Clinical Trial.
The end of the recruitment of subjects for the clinical trial and the end of the clinical trial should be notified in CTIS. That notification shall be made within 15 days from the end of the clinical trial in relation to that Member State.

13. Summary of the results of the clinical trial

The end of the Clinical Trial corresponds to the end of the participation of the last person who participate to the Clinical Trial.
The end of the recruitment of subjects for the clinical trial and the end of the clinical trial should be notified in CTIS. That notification shall be made within 15 days from the end of the clinical trial in relation to that Member State

14. Archiving

Specific documents for a clinical trial on medicinal product for human use will be archived by the investigator and the sponsor for 25 years after the end of the research.

This indexed archiving includes, in particular:

- A sealed envelope for the investigator containing a copy of all the information notes and consent forms signed by all individuals at the centre who participated in the study;
- A sealed envelope for the sponsor containing a copy of all the information notes and consent forms signed by all individuals at the centre who participated in the study;
- "Study" binders for the Investigator and the sponsor, including (non-exhaustive list):

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- the successive versions of the protocol (identified by the version number and its date), and any appendices
- the ANSM authorisations and CPP (Research Ethics Committee) decisions
- any correspondence
- the enrolment list or register
- the appendices specific to the research
- final study report

- The data collection documents

15 FUNDING AND INSURANCE

15.1 Funding sources

This research has been founded by Institut National du Cancer (INCa, PHRC-K18-148), the Ligue Nationale Contre le Cancer, and by Miltenyi Biotec, which provided the lentiviral vector.

15.2 Insurance

For the duration of the study, the Sponsor will take out an insurance policy covering the sponsor's own public liability, as well as the public liability for all the physicians involved in the study. The sponsor will also provide full compensation for any damages caused by the study to the participant enrolled and their beneficiaries, unless the sponsor can prove that the harm is not the fault of the sponsor or any collaborator. Compensation cannot be refused on the grounds of a third-party act or the voluntary withdrawal of the person who initially consented to participate in the study.

Assistance Publique-Hôpitaux de Paris (AP-HP) has taken out insurance with AXA XL through the insurance broker HOWDEN France for the full study period, which covers its own public liability and that of any collaborator (physician or research staff), in accordance with with Article 76(1) of regulation (EU) No 536/2014 of the European Parliament and of the council of 16 April 2014 and Article L.1121-10 of the *Code de la Santé Publique* (French Public Health Code).

16 PUBLICATION RULES

The author(s) of any publication relating to this study must include the AP-HP among their affiliations and must name the sponsor AP-HP (DRCI) and the source of funding (INCa, Miltenyi Biotec). A copy of the publication must be sent to the sponsor.

16.1 Mention of AP-HP affiliation for projects sponsored by AP-HP

If an author has several affiliations, the order in which the institutions are mentioned (AP-HP, University, INSERM, etc.) is unimportant
Each of these affiliations must be identified by an address separated by a semicolon (;)
The AP-HP institution must feature under the acronym "**AP-HP**" first in the address, specifically followed by: **AP-HP**, hospital, department, city, postcode, France

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16.2 Mention of the sponsor AP-HP (DRCI) in the acknowledgements of the text

"The sponsor was Assistance Publique – Hôpitaux de Paris (Délégation à la Recherche Clinique et à l'Innovation)"

16.3 Mention of the financial backer in the acknowledgements of the text

"The study was funded by a grant from Programme Hospitalier de Recherche Clinique - PHRC 2018 (French Ministry of Health)".

This study has been registered on the website http://clinicaltrials.gov/ under number *(add the registration number when the study is registered)*.

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LIST OF ADDENDA

Each addendum and the log of addenda versions are attached, independently of the protocol. Each addendum can be modified (change of addendum version) without modifying the protocol version.

17.1 List of investigators

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17.2 List of establishments responsible for activities concerning advanced therapy medicinal products

ATMP platform (MEARY, Saint-Louis, APHP, Paris)		Fabreguettes, Jean Roch	jean-roch.fabreguettes@aphp.fr

17.3 Serious Adverse Events notification form

17.4 Pregnancy notification form

17.5 Investigator's Brochure

17.6 Questionnaire or scale

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